# Chapter 18 Electromagnetic Induction

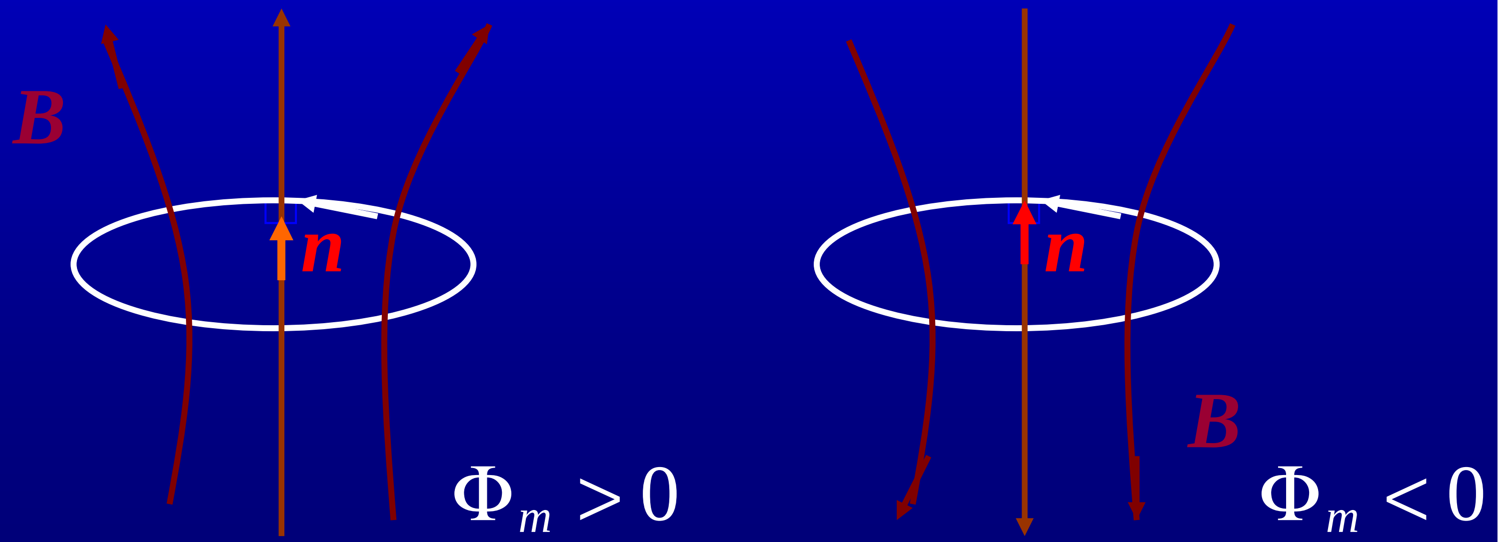
## 18.1 Faraday's law of induction

**Faraday's law of induction**

$$\varepsilon_i = -\frac{d\Phi_m}{dt}.$$

induced electromotive force

right-hand rule



A non-electrostatic field is induced when a magnetic field is changing with time.

static electric field

induced electric field

$$\oint_L \boldsymbol{E}_s \cdot d\boldsymbol{l} = 0. \qquad \oint_L \mathbf{E}_i \cdot d\mathbf{l} = \varepsilon_i .$$

$$\Downarrow$$

$$\varepsilon_i = \oint_L \mathbf{E} \cdot d\mathbf{l}, \qquad \mathbf{E} = \mathbf{E}_i + \mathbf{E}_s .$$

$$\Downarrow$$

$$\boxed{\oint_L \boldsymbol{E} \cdot d\boldsymbol{l} = -\frac{d}{dt}\int_S \boldsymbol{B} \cdot d\boldsymbol{S}.}$$

## *Faraday's law in differential form

Using Stokes' theorem

$$\oint_L \boldsymbol{E} \cdot d\boldsymbol{l} = \int_S \nabla \times \boldsymbol{E} \cdot d\boldsymbol{S},$$

Faraday's law can be rewritten as

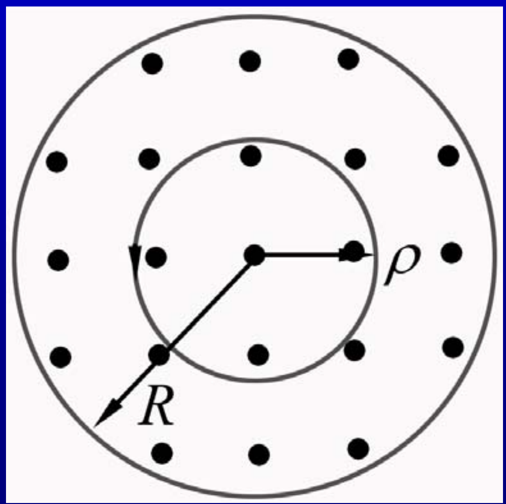
$$\int_S \nabla \times \boldsymbol{E} \cdot d\boldsymbol{S} = -\int_S \frac{\partial \boldsymbol{B}}{\partial t} \cdot d\boldsymbol{S}.$$

$$\Downarrow$$

$$\boxed{\nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}.}$$

**Example 18.1** Determine the electric field produced by an alternating axial magnetic field.

**Solution:** The symmetry of the problem suggests that the lines of force of the induced field are circles concentric with the axis. Choose a path *L* which is a circle of radius $\rho$ concentric with the axis. Using Faraday's law, we have




$$\oint_L \boldsymbol{E} \cdot d\boldsymbol{l} = E_\varphi (2\pi\rho)$$

$$= -\frac{d}{dt}\int_0^\rho B(\rho) 2\pi\rho d\rho.$$

Hence

$$E_\varphi = -\frac{1}{\rho}\int_0^\rho \frac{\partial B(\rho)}{\partial t}\rho d\rho.$$

Defining

$$B_{\mathrm{ave}}(\rho) = -\frac{1}{\pi\rho^2}\int_0^\rho B(\rho)2\pi\rho d\rho,$$

the above expression for the induced field can be rewritten as

$$E_\varphi = -\frac{\rho}{2}\frac{\partial B_{\mathrm{ave}}(\rho)}{\partial t}.$$

Particularly, if

$$B(\rho) = \begin{cases} B, & \rho < R, \\ 0, & \rho > R, \end{cases}$$

we have

$$E_\varphi(\rho) = \begin{cases} -\dfrac{\rho}{2}\dfrac{dB}{dt}, & \rho < R, \\ -\dfrac{R^2}{2\rho}\dfrac{dB}{dt}, & \rho > R. \end{cases}$$

**Example 18.2** Electirc field of a pulsating solenoid

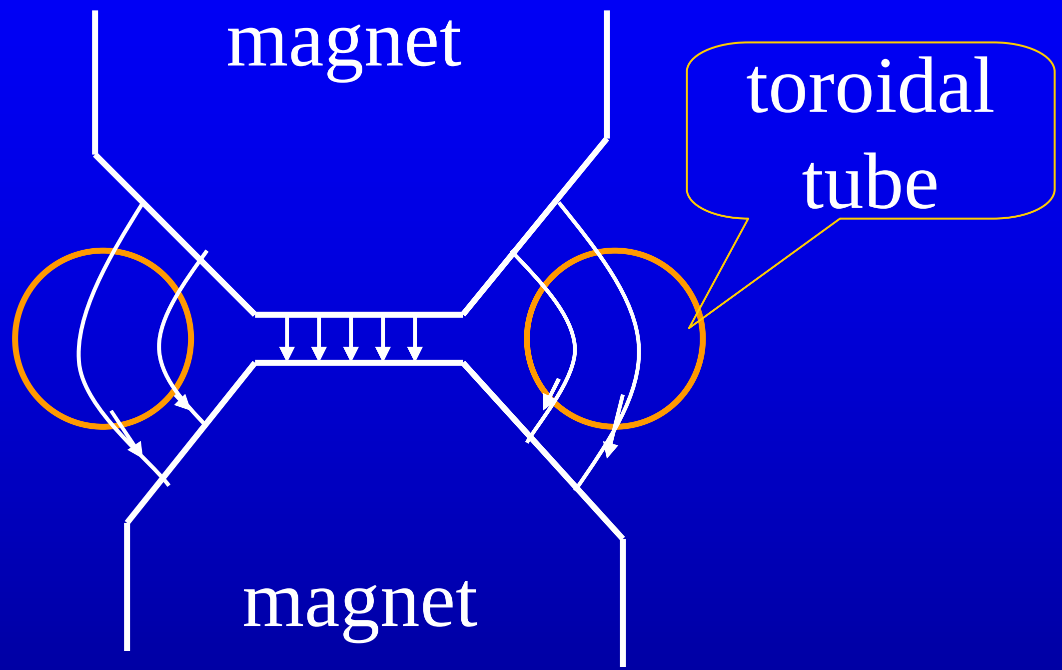
**Solution:** The magnetic field of the solenoid is

$$B_{\text{in}} = \mu_0 nI(t), \qquad B_{\text{out}} = 0.$$

Therefore, using the result of above example, we have

$$E_\varphi(\rho) = \begin{cases} -\dfrac{\rho\mu_0 n}{2}\dfrac{dI(t)}{dt}, & \rho < R, \\ -\dfrac{R^2\mu_0 n}{2\rho}\dfrac{dI(t)}{dt}, & \rho > R. \end{cases}$$

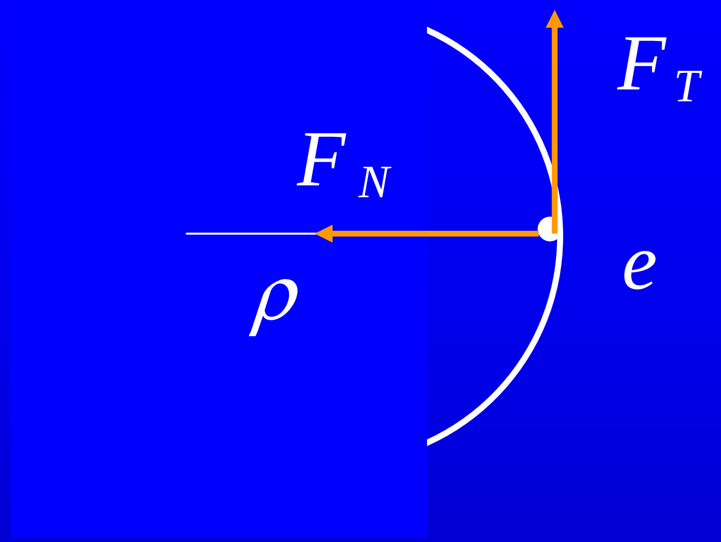
# **Betatron** is an electron accelerator



$$F_T = -eE = \frac{e\rho}{2}\frac{\partial B_{\mathrm{ave}}(\rho)}{\partial t},$$

$$F_N = evB(\rho).$$

$$m\frac{v^2}{\rho} = p\frac{v}{\rho} = evB(\rho).$$




$$\Downarrow$$

$$p = e\rho B(\rho).$$

$$\frac{dp}{dt} = e\rho\frac{dB(\rho)}{dt}, \qquad \frac{dp}{dt} = F_T = \frac{e\rho}{2}\frac{\partial B_{\mathrm{ave}}(\rho)}{\partial t},$$

$$B(\rho) = \frac{1}{2}B_{\mathrm{ave}}(\rho).$$

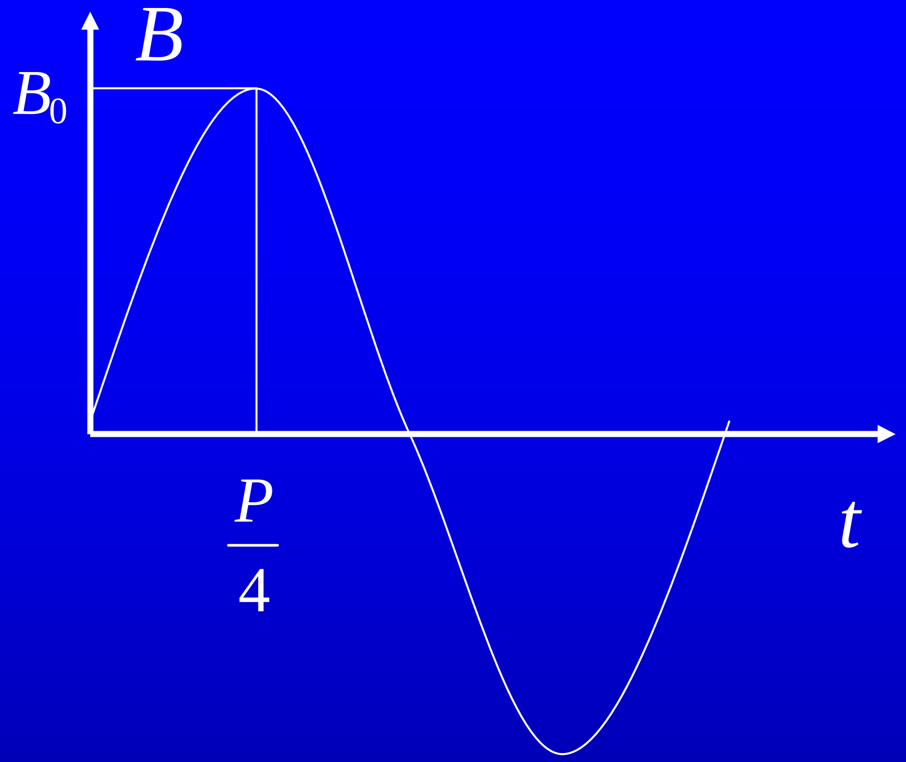
condition for stable cyclotron motion

$$\frac{dp}{dt} = e\rho \frac{dB(\rho)}{dt}.$$

$$\int_0^{P/4} \frac{dp}{dt} dt = e\rho \int_0^{B_0} \frac{dB(\rho)}{dt} dt.$$

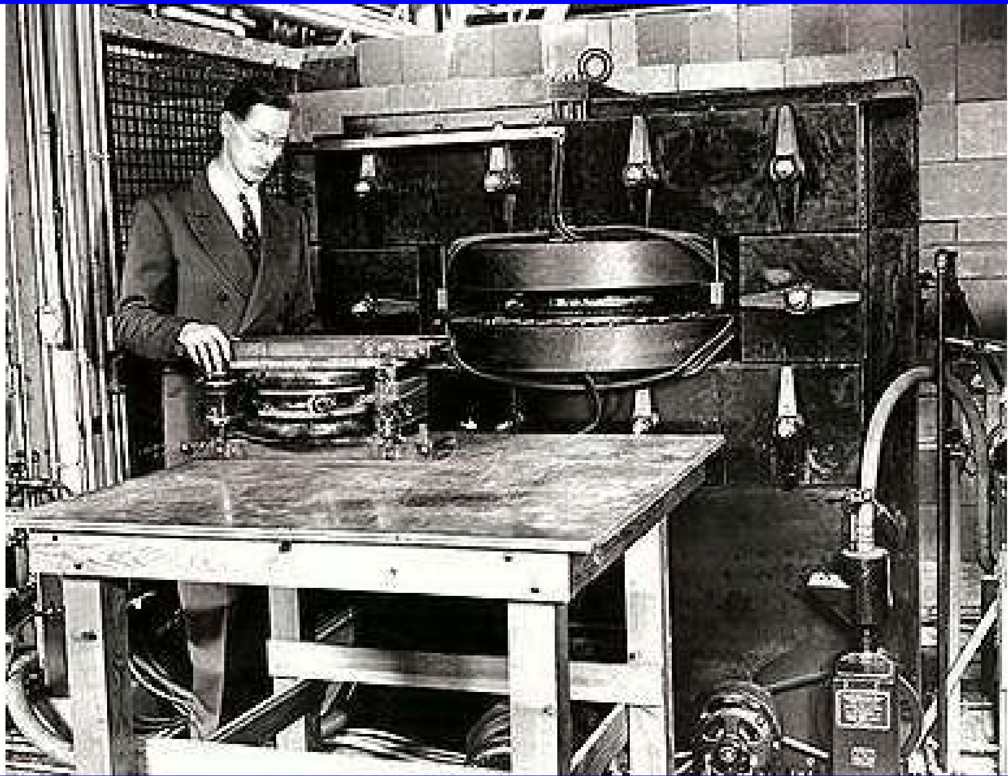
$$p_{\max} = e\rho B_0.$$

$$T_{\max} = \frac{p_{\max}^2}{2m_e} = \frac{e^2 \rho^2 B_0^2}{2m_e}.$$




350MeV

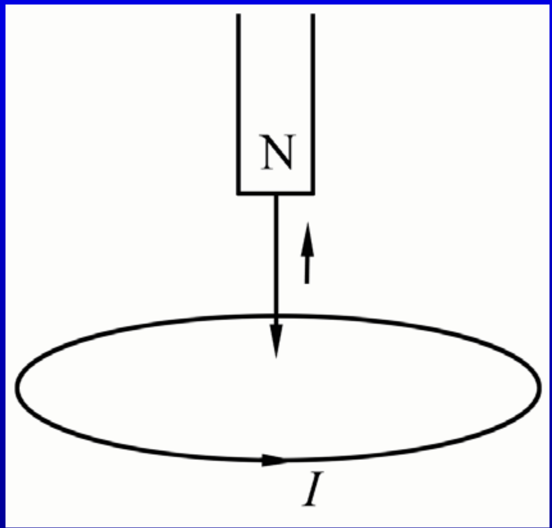
nuclear reactions
medical treatments



**Kerst, Donald William** (1911-93), U.S. physicist, born in Galena, Ill.; joined faculty of University of Illinois 1938, professor of physics 1943-57; invented betatron there.

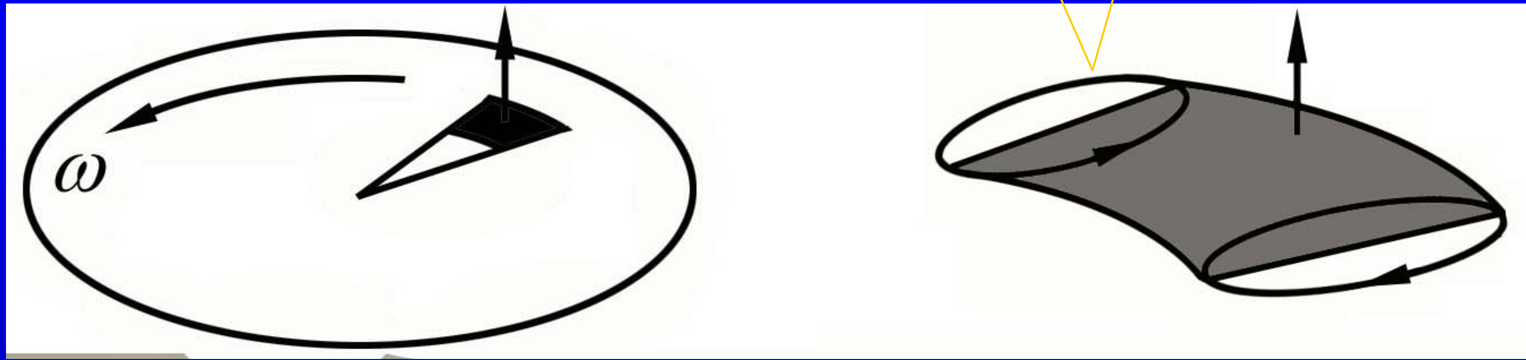
## Lenz' law

The induced current produces flux which resists the original change in flux




a conducting sheet can shield varying magnetic field leaving the square

eddy current brake



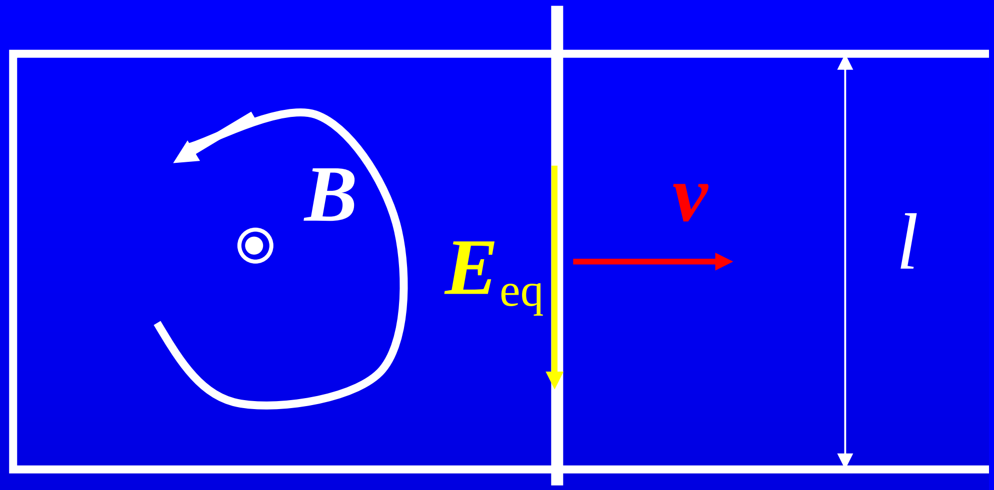

$$\varepsilon \sim B\frac{\mathrm{d}S}{\mathrm{d}t} = B\frac{r\,\mathrm{d}\varphi\, a}{\mathrm{d}t} = Bra\omega$$

$$I = jat = \sigma Eat = \sigma\varepsilon t$$

$$M = rF = r(Ia)B = B^2a^2r^2\omega\sigma t$$

laminated core

## 18.2 Motional emf




$$F = q\boldsymbol{v}\times\boldsymbol{B}.$$

an 'equivalent' electric field $\boldsymbol{E}_{eq} = \boldsymbol{v}\times\boldsymbol{B}.$

non-electrostatic field

$$E_{eq} = -vB. \quad \varepsilon_i = -vBl.$$

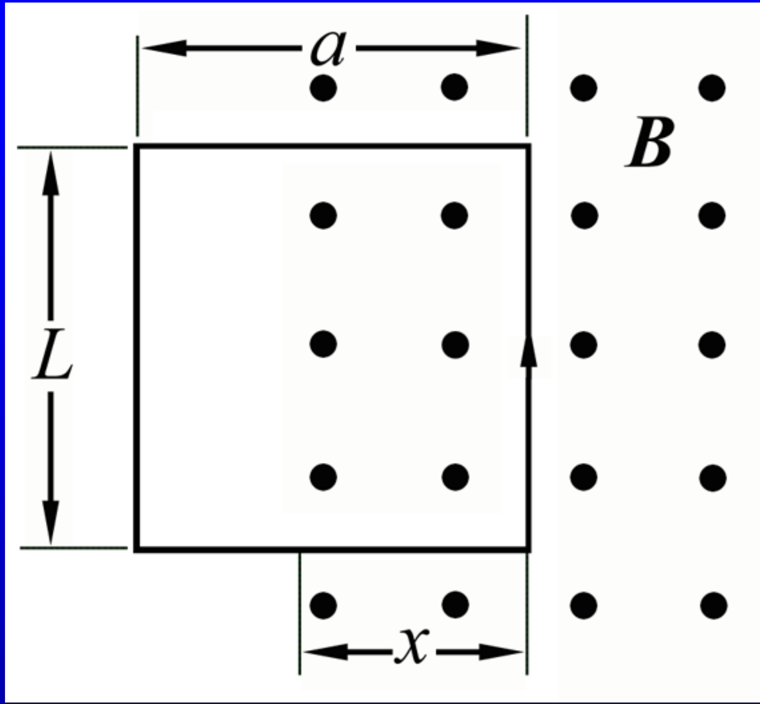
$$\Phi_m = Blx, \; -\frac{d\Phi_m}{dt} = -Bl\frac{dx}{dt} = -Blv. \Longrightarrow \varepsilon_i = -\frac{d\Phi_m}{dt}.$$

The Faraday's law of electromagnetic induction

$\varepsilon_i = -d\Phi_m / dt$ can be applied when the change in magnetic flux is due either to a change in the magnetic field or to a motion or deformation of a circuit along which the emf is calculated, or both.

The induced emf due to the motion---motional emf

## **Example 18.3** Induction in moving loop




$$\Phi = \begin{cases} BLx & (a > x > 0) \\ BLa & (x > a) \end{cases}$$

$$\varepsilon = -\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

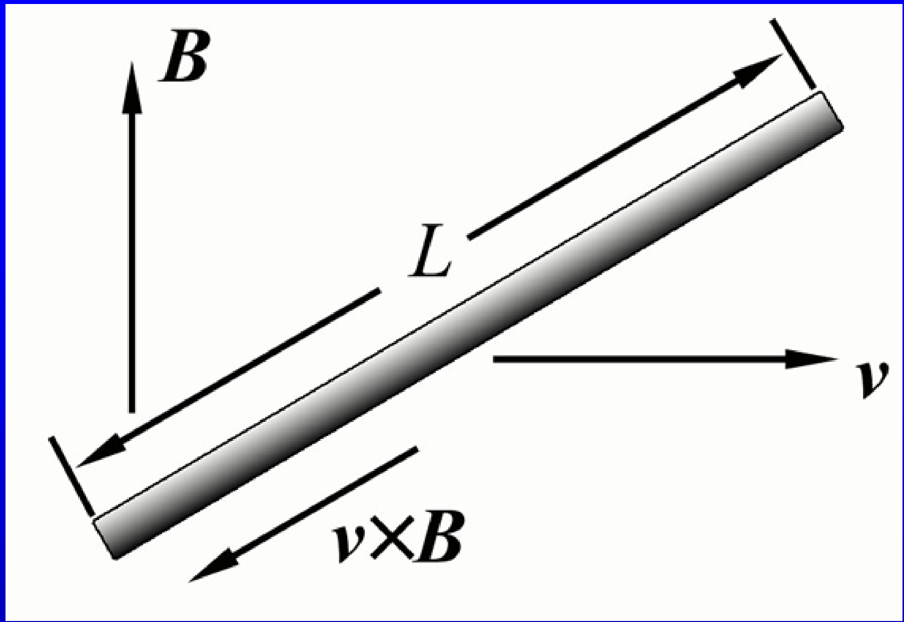
$$= -BLv$$

$$I = \frac{|\varepsilon|}{R} = \frac{BLv}{R}$$

$$F = ILB = \frac{B^2L^2v}{R}$$

$$P = Fv = \frac{B^2L^2v^2}{R}, \quad P_Q = I\varepsilon = I^2R = \frac{B^2L^2v^2}{R}$$

## **Example 18.4** Moving bar




transient process:
charge accumulation
forming $E$,
keeps increasing until

$$\boldsymbol{F} = q\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right) \equiv q(\boldsymbol{E} + \boldsymbol{E}_{\mathrm{eq}}) = \mathbf{0}.$$

Induced electric field and emf are

$$\mathbf{E}_{\mathrm{i}} = \mathbf{E}_{\mathrm{eq}} = \mathbf{v} \times \mathbf{B},$$

$$\mathcal{E}_{\mathrm{i}} = \int \mathbf{E}_{\mathrm{i}} \cdot d\mathbf{l} = vBL.$$

In bar frame observer sees

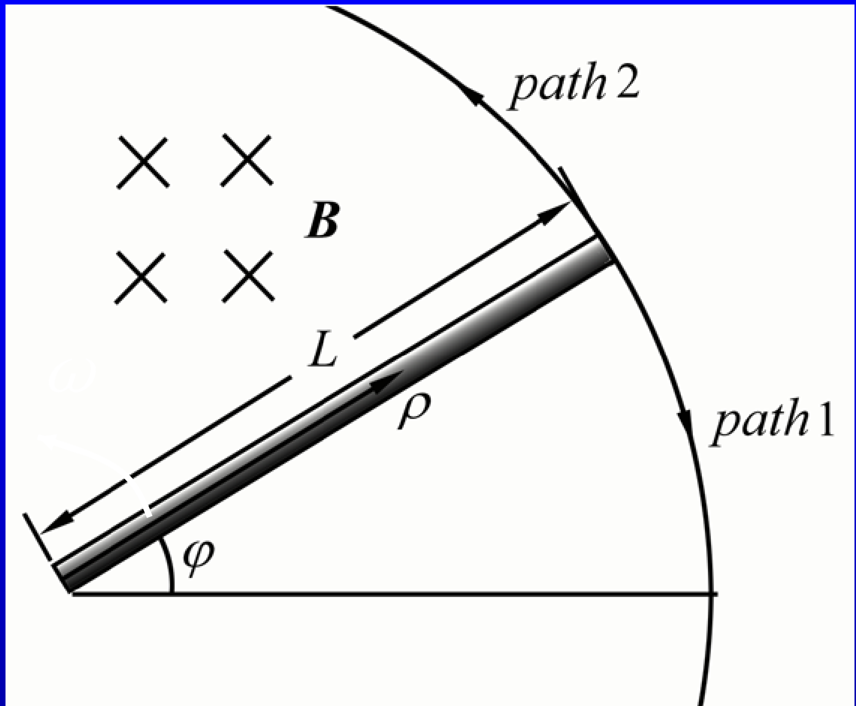
$$\boldsymbol{B}' = \gamma \boldsymbol{B} \approx \boldsymbol{B},$$

$$\boldsymbol{E}' = \gamma \boldsymbol{v} \times \boldsymbol{B} \approx \boldsymbol{v} \times \boldsymbol{B}.$$

***E'*** that causes the redistribution of charges forming ***E***

$\boldsymbol{E}' + \boldsymbol{E} = \mathbf{0}.$

## **Example 18.5** Rotating bar




Assume $\boldsymbol{\omega} = \omega\, \boldsymbol{e}_z$, $\boldsymbol{B} = -B\, \boldsymbol{e}_z$,

$$\boldsymbol{E}_{\mathrm{i}}(\rho) = \boldsymbol{v}\times\boldsymbol{B}$$

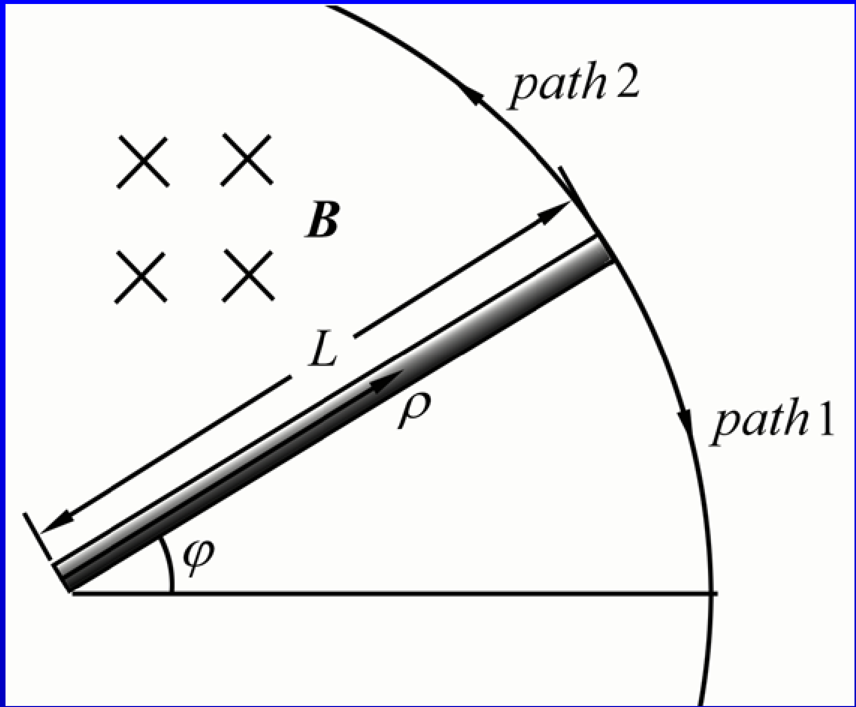
$$= (\boldsymbol{\omega}\times\boldsymbol{\rho})\times\boldsymbol{B}$$

$$= -B\omega\rho\,(\boldsymbol{e}_z\times\boldsymbol{e}_\rho)\times\boldsymbol{e}_z$$

$$= -B\omega\rho\ \boldsymbol{e}_\varphi\times\boldsymbol{e}_z$$

$$= -B\omega\rho\boldsymbol{e}_\rho = -B\omega\boldsymbol{\rho}\,.$$

$$\varepsilon_{\mathrm{i}} = \int_0^L \mathbf{E}_{\mathrm{i}}\cdot d\mathbf{l} = \int_0^L - B\omega\boldsymbol{\rho}\cdot d\boldsymbol{\rho} = \int_0^L - B\omega\rho\ d\rho = -\frac{1}{2}\omega BL^2.$$




For path 1. We can get flux and emf

$$\Phi_m = \frac{1}{2}(L\varphi \cdot L)B > 0$$

$$\dot{\Phi}_m = \frac{1}{2}BL^2\dot{\varphi},$$

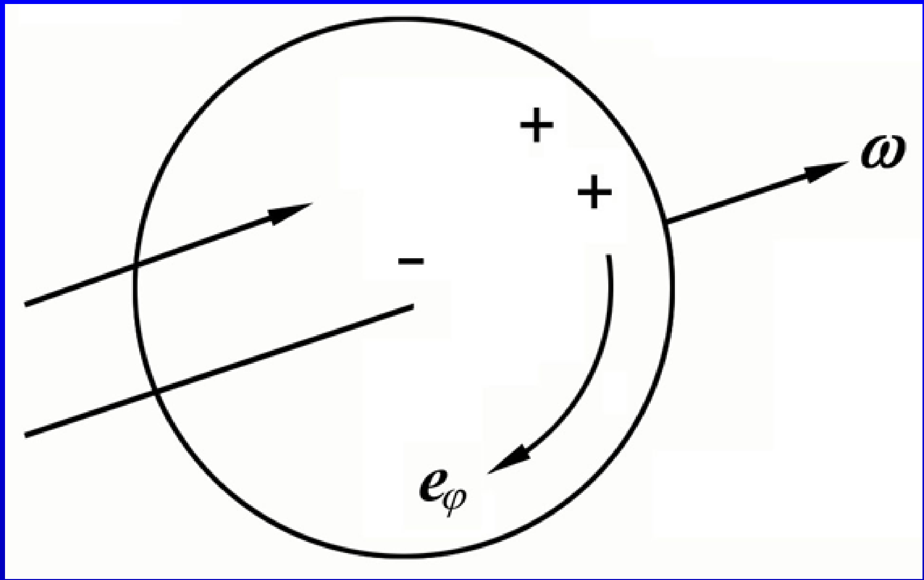
$$\mathrm{E}_\mathrm{i} = -\dot{\Phi}_m = -\frac{1}{2}BL^2\omega.$$

Alternatively we can chose another loop, path 2

$$\Phi_m = -\left(\pi L^2 - \frac{1}{2}\varphi L^2\right)B < 0$$

$$\dot{\Phi}_m = \frac{1}{2}BL^2\dot{\varphi}, \qquad \varepsilon_\mathrm{i} = -\dot{\Phi}_m = -\frac{1}{2}BL^2\omega.$$

**Example 18.6** Faraday's disc (induction generator)




$$\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{\rho} = \omega\rho\, \boldsymbol{e}_\varphi.$$

$$\boldsymbol{E}_\mathrm{i} = \boldsymbol{v} \times \boldsymbol{B} = \omega B \rho$$

$$\varepsilon_\mathrm{i} = \frac{1}{2}\omega B R^2$$

For $f = 1000\mathrm{Hz}, B = 0.1\mathrm{T},\ R = 0.2\mathrm{m}.$

the emf can be 12V

# Electromagnetic induction and the principle of the relativity

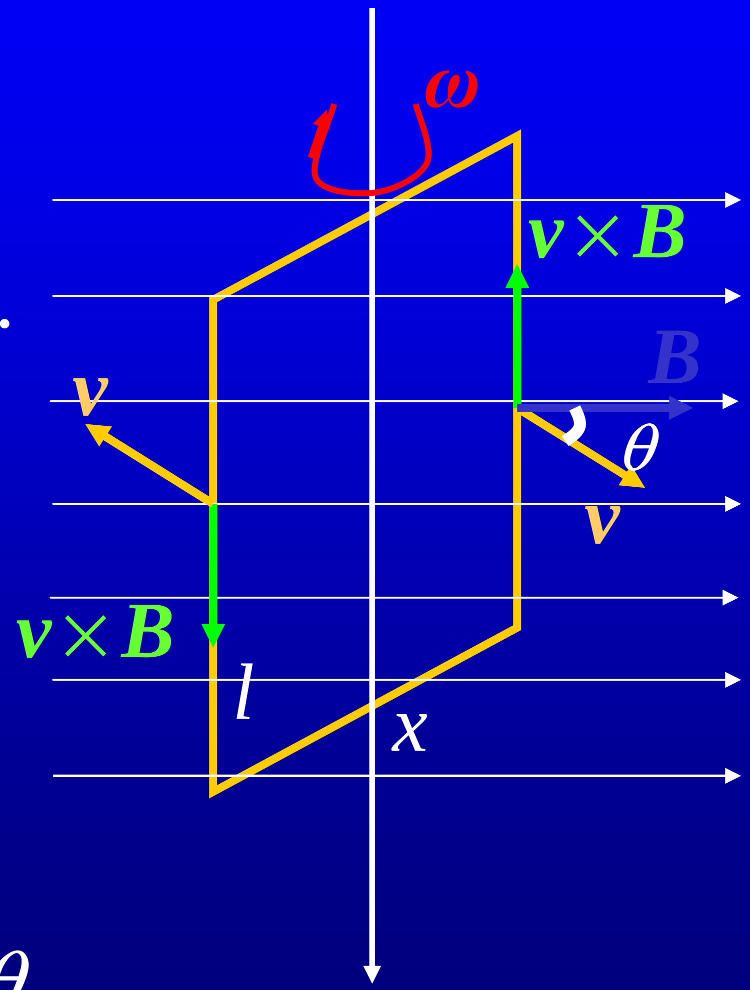
$$\varepsilon_{\mathrm{i}} = \oint \mathbf{E}_{\mathrm{eq}} \cdot d\mathbf{l} = 2lvB\sin\theta$$

$$= 2l\omega\left(\frac{x}{2}\right)B\sin\theta = BS\omega\sin\theta.$$

$$\theta = \omega t, \quad \Phi_m = BS\cos\omega t.$$

$$\varepsilon_{\mathrm{i}} = -\frac{d\Phi_m}{dt} = \omega BS\sin\omega t.$$

$$= 2l\omega\left(\frac{x}{2}\right)B\sin\theta = BS\omega\sin\theta.$$



the rest observer:

•no electric field

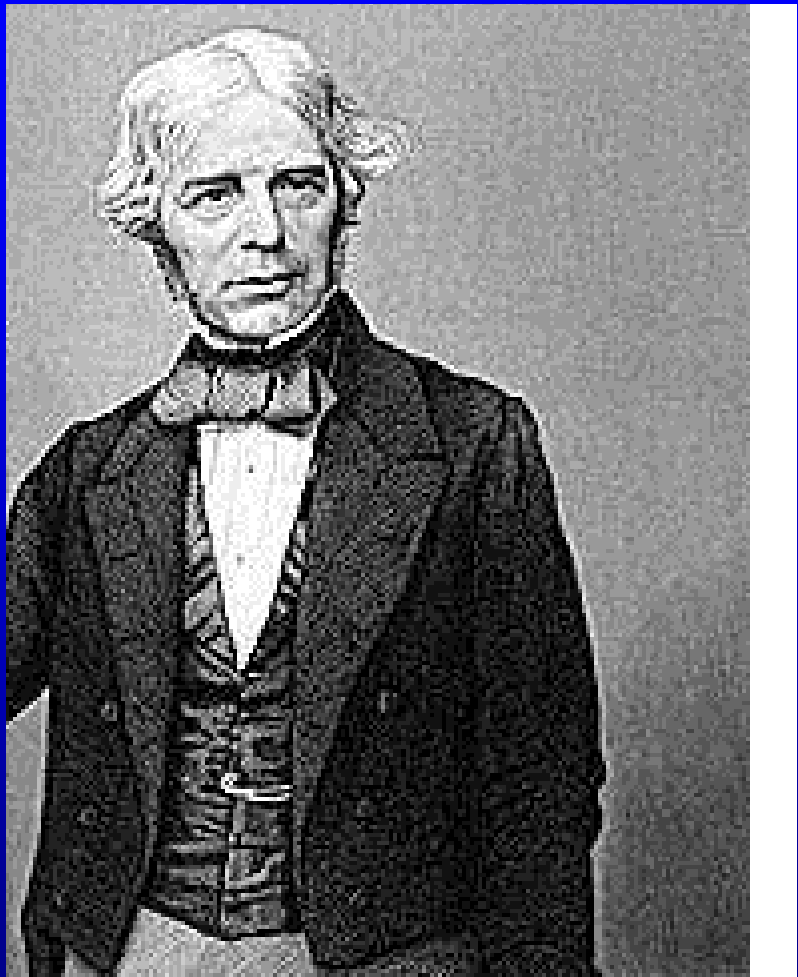
•the emf is due to the magnetic force

The observer attached to the rotating frame:

•the direction of the magnetic field is rotating in the space $\frac{\partial \boldsymbol{B}}{\partial t} = -\boldsymbol{\omega} \times \boldsymbol{B}.$

•the force acted on the electron is due to electric field associated with the changing magnetic field

Assignment :18.1, 18.2



**FARADAY, Michael** (1791-1867). The English physicist and chemist Michael Faraday made many notable contributions to chemistry and electricity. When the great scientist Sir Humphry Davy was asked what he considered his greatest discovery, he answered, "Michael Faraday."

Faraday was born in Newington, Surrey, England, on Sept. 22, 1791. The son of a blacksmith, he was apprenticed to a bookbinder at age 14 and read all the scientific books in the shop. Young Faraday attended lectures by Sir Humphry Davy. He made careful notes and sent them to Davy, asking for a job. Impressed by the boy's zeal, the scientist took Faraday into his laboratory as an assistant.

Acting on hints from Davy, he succeeded in liquefying gas by compression. When he discovered the hydrocarbon benzene in 1825, he became the father of an entire branch of organic chemistry. His laws of electrolysis, formulated in 1833, linked chemistry and electricity.

Faraday's greatest achievement was the discovery of electromagnetic induction. He found in 1831 that when he moved a magnet through a coil of wire, a current was produced. From this discovery the electric generator--the heart of all modern electric power plants--was developed.

Late in his career Faraday discovered the rotation of the plane of polarization of light in a strong magnetic field His work in electromagnetism led James Clerk Maxwell to the theory that linked electricity, magnetism, and light. Faraday died at Hampton Court, Surrey, on Aug. 25, 1867.

Royal Institution : Thursday evening.
(December, l820)

My dear Sarah-It is astonishing how much the state of the body influences the powers of the mind. I have been thinking all the morning the very delightful and interesting letter I would send you this evening, and now I am so tired, and yet have so much to do that my thoughts are quite giddy, and run round your image without any power of themselves to stop and admire it. I want to say a thousand kind and, believe me, heartfelt thing to you, but am not master of words fit for the purpose; and still, as I ponder and think on you, chlorides, trials, oil, Davy, steel, miscellanea, mercury, and fifty other professional fancies swim before and drive me further and further into the quandary of stupidness.

From your affectionate

Michael

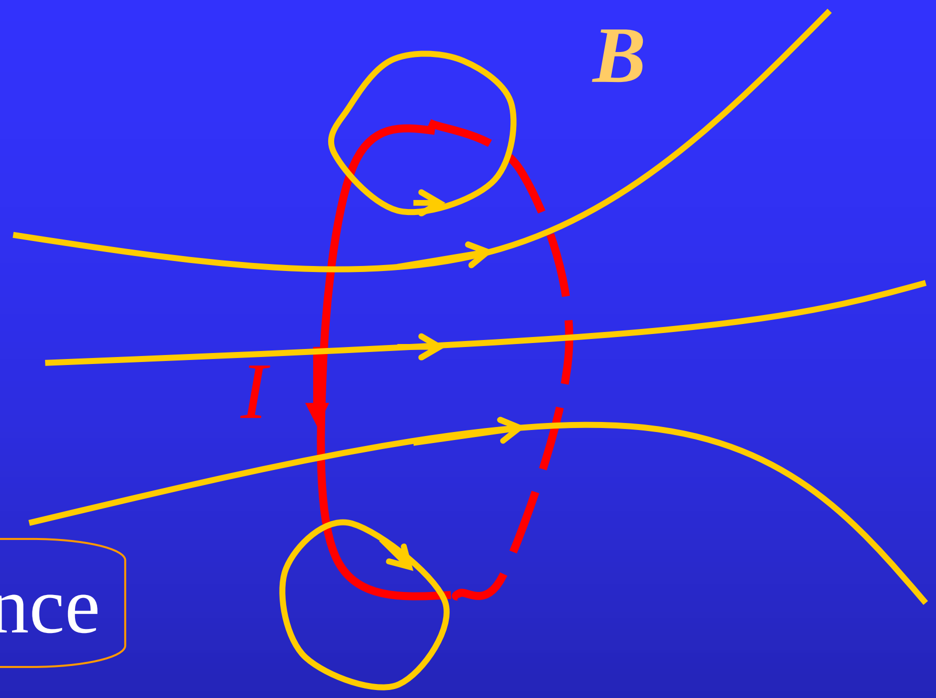
## 18.3 Inductance

### Self-inductance

$$B \propto I \Rightarrow \Phi_m \propto I$$

$$\Phi_m = LI$$

self -inductance




unit: $WbA^{-1}$, henry (H)

the self-flux $\Phi_m$—the flux through the circuit produced by its own current

$$q = CV$$

capacitance

the change in the current $\longrightarrow$ the change in the self-flux $\longrightarrow$ induced emf—<u>self induction</u>

Differentiating both sides of $\Phi_m = LI$ gives

$$\varepsilon_i = -\frac{d\Phi_m}{dt} = -L\frac{dI}{dt}.$$

$$L = -\frac{\varepsilon_i}{\frac{dI}{dt}}.$$

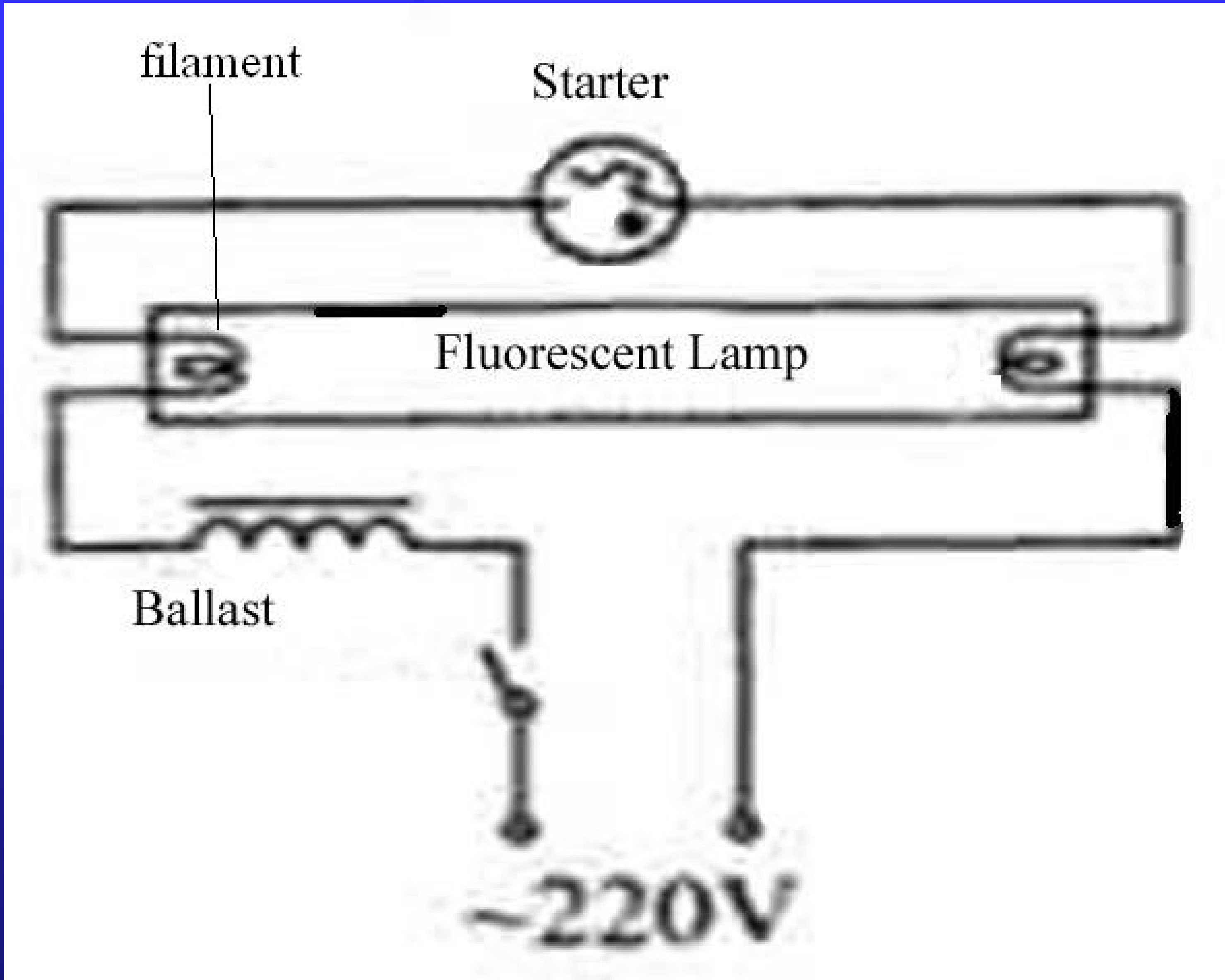
In experiments, self-inductance can be measured as ratio of the induced emf to the changing rate of the current.

—<u>operational definition.</u>

When $dI/dt > 0$, $\varepsilon_i$ is opposed to the current.

When $dI/dt < 0$, $\varepsilon_i$ acts in the same direction of the current..

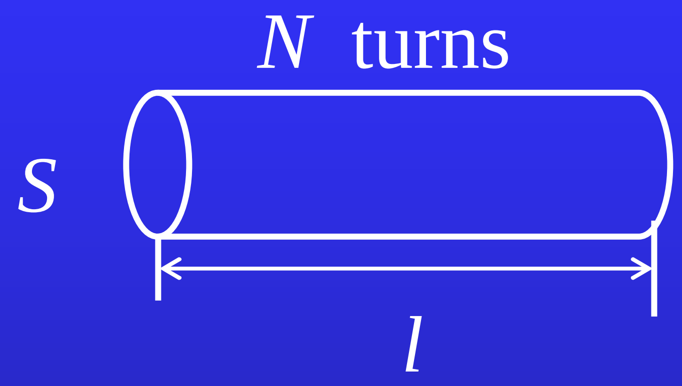
# Ballast for the fluorescent lamp



## The self-inductance of a solenoid

For a solenoid of $N$ turns , the self-flux is

$$\Phi_m = N\Phi_1 = NBS.$$




$$B = \mu_0 nI \qquad (n = N/l).$$

$$\Phi_m = N\mu_0 nIS = \mu_0 n^2 lSI.$$

$$\Downarrow$$

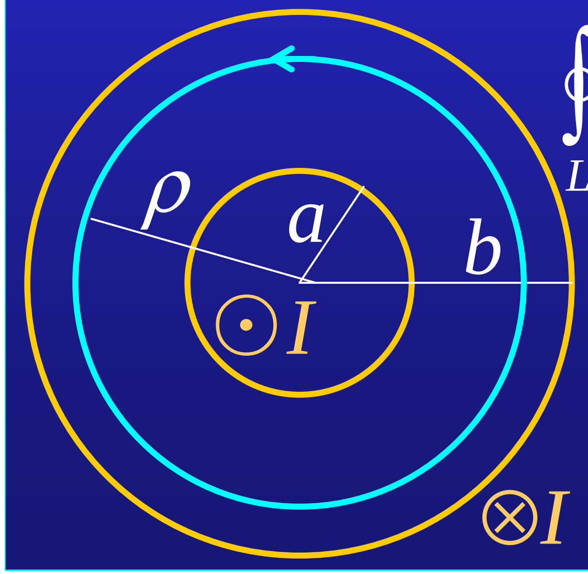
$$L = \mu_0 n^2 lS = \mu_0 n^2 V.$$

volume of the solenoid

**Example 18.7** Find the self-inductance of a coaxial cable.

**Solution:** Suppose the two coaxial, cylindrical metal shells of the cable are of radii a and b respectively. Each shell carries a current I, but in opposite direction. The magnetic field in the region between two two shells can be computed as

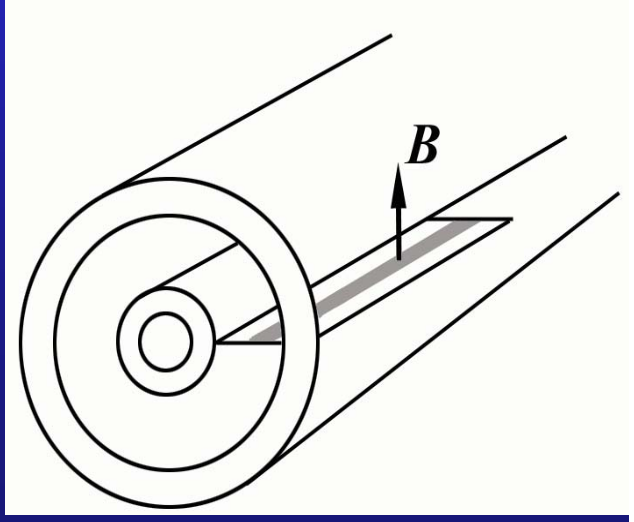
$$B = \frac{\mu_0 I}{2\pi\rho}.$$




$$\oint_L \boldsymbol{B} \cdot d\boldsymbol{l} = B \cdot 2\pi\rho = \frac{I}{\mu_0}.$$

$$\Downarrow$$

$$B = \frac{\mu_0 I}{2\pi\rho}.$$

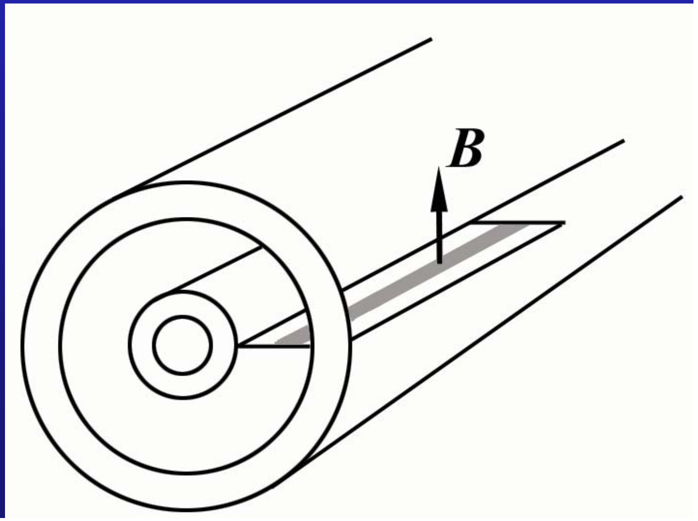


Then, the magnetic flux through the section of length $l$ (as shown in the figure) is

$$\Phi = \int_a^b \frac{\mu_0 I}{2\pi\rho} l d\rho = \frac{\mu_0 I l}{2\pi} \ln\frac{b}{a}.$$

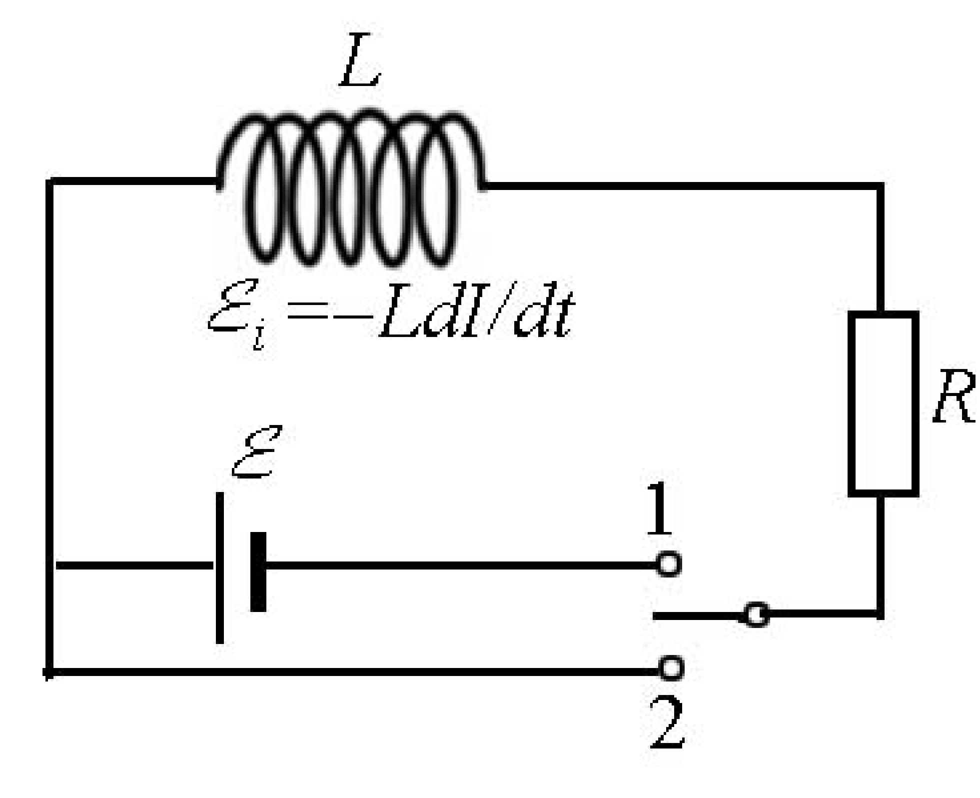
Therefore the self-inductance for a portion of the cable of length $l$ is

$$L = \frac{\mu_0 l}{2\pi} \ln\frac{b}{a}.$$



**Example:** Establishment and decay of a current in a circuit with self-inductance.

**Solution:** When an emf is applied by closing the switch to position 1, the current will increase gradually in the circuit. Because of the existence of the self-inductance, the self-induced emf $\varepsilon_i$ will appear. Thus, using Ohm's law, we have




$$\varepsilon + \varepsilon_i = RI, \quad \text{or} \quad \varepsilon - L\frac{dI}{dt} = RI. \tag{1}$$

Eq. (1) can be rewritten as

$$\frac{dI}{I-\dfrac{\varepsilon}{R}}=-\frac{R}{L}dt. \tag{2}$$

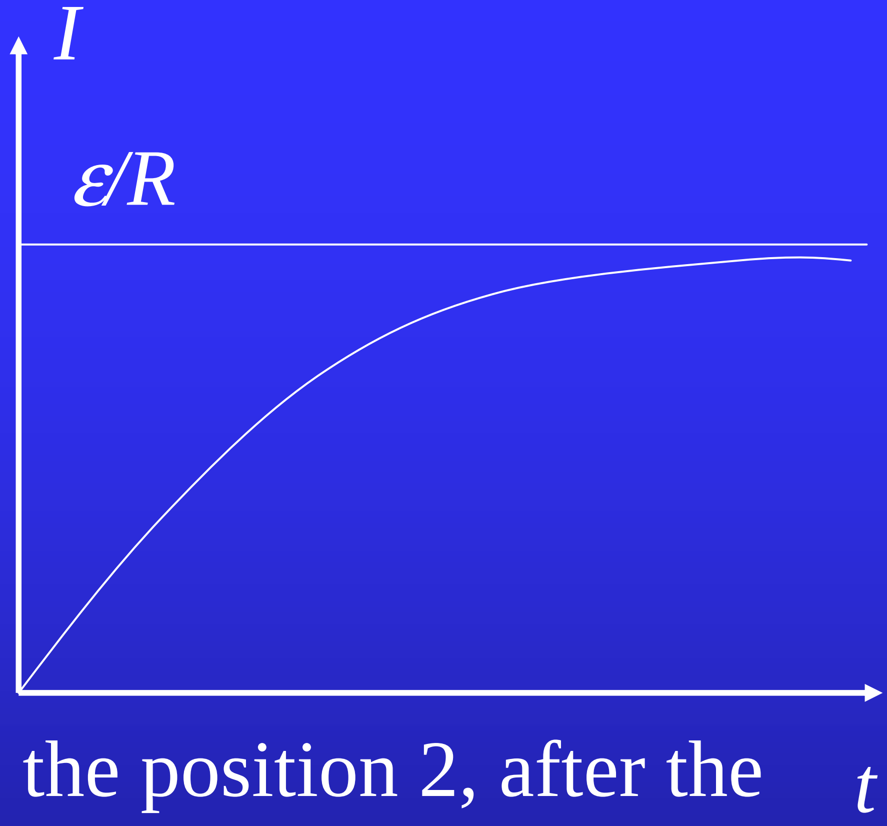
Noting that at $t$=0, the current is zero, then

$$\int_0^I \frac{dI}{I-\dfrac{\varepsilon}{R}}=-\int_0^t \frac{R}{L}dt, \tag{3}$$

and the integration gives

$$\ln\frac{I-\dfrac{\varepsilon}{R}}{-\dfrac{\varepsilon}{R}}=-\frac{R}{L}t, \qquad \text{or} \qquad I=\frac{\varepsilon}{R}(1-e^{-\frac{R}{L}t}). \tag{4}$$

When $t \to \infty$, the current achieves its limiting value $\varepsilon / R$.




If we move the switch over to the position 2, after the current reaches its steady value $\varepsilon / R$, the applied emf is removed. The Ohm's law then become

$$-L\frac{dI}{dt} = RI \quad \text{or} \quad \frac{dI}{I} = -\frac{R}{L}dt. \qquad (4)$$

Noting that the initial current is $\varepsilon/R$, the integration of Eq. (4) gives

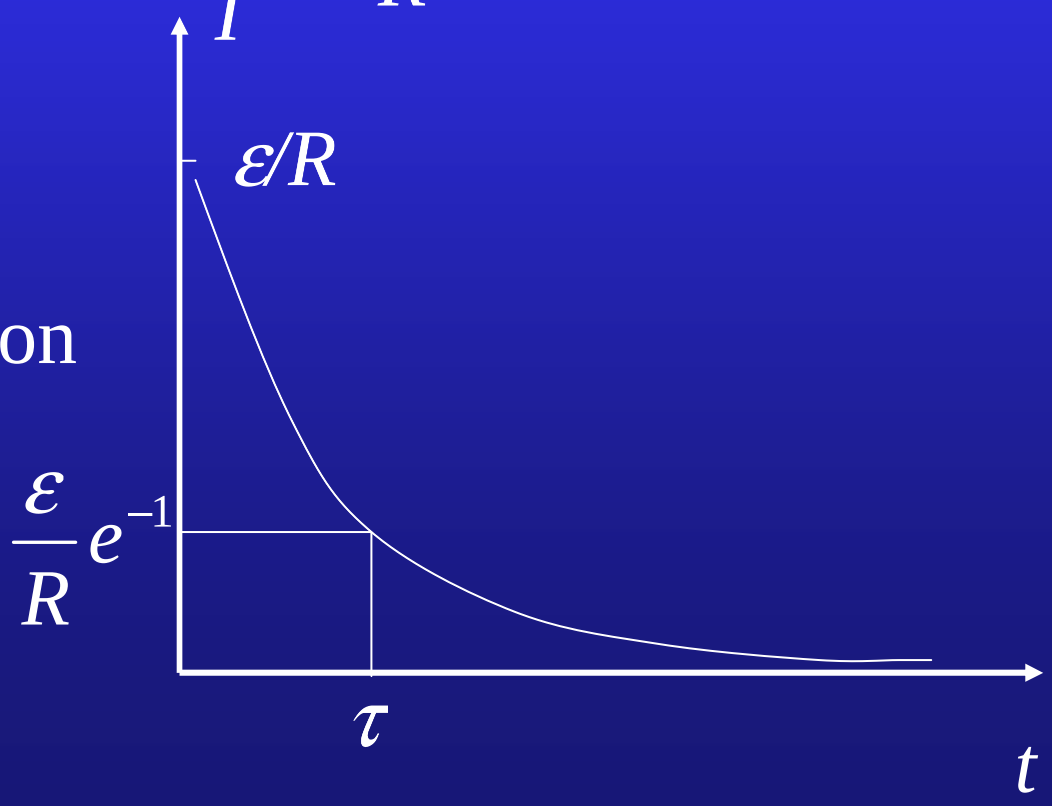
$$\int_{\varepsilon/R}^{I}\frac{dI}{I}=-\frac{R}{L}\int_0^t dt \quad \text{or} \quad \ln\frac{I}{\frac{\varepsilon}{R}}=-\frac{R}{L}t \qquad (5)$$

or

$$I=\frac{\varepsilon}{R}e^{-\frac{R}{L}t}. \qquad (6)$$

$$\tau=\frac{L}{R}$$ ---the relaxation time

$$I=\frac{\varepsilon}{R}e^{-t/\tau}.$$



## Energy of magnetic field

$$\varepsilon = RI + L\frac{dI}{dt},$$

$$\varepsilon I = RI^2 + LI\frac{dI}{dt}.$$

the work done by the battery per unit time

the Joule heat power

the energy required to build up the current per unit time

⇕

the energy needed to establish the magnetic field per unit time

Therefore, the increasing rate of the magnetic energy is

$$\frac{dU_m}{dt} = LI\frac{dI}{dt}.$$

The magnetic energy stored in the inductor is

$$U_m = \int_0^t LI\frac{dI}{dt}dt = \int_0^t LIdI = \frac{1}{2}LI^2.$$

For a solenoid, $L = \mu_0 n^2 lS, \quad B = \mu_0 nI.$

$$U_m = \frac{1}{2}\mu_0 n^2 lS\frac{B^2}{(\mu_0 n)^2} = \frac{1}{2\mu_0}B^2 V.$$

$$\boxed{u_{\mathrm{m}} = \frac{1}{2\mu_0}\boldsymbol{B}\cdot\boldsymbol{B}.}$$

For a coaxial cable

$$L = \frac{\mu_0 l}{2\pi} \ln \frac{b}{a},$$

$$U_m = \frac{1}{2} L I^2 = \frac{\mu_0 l I^2}{4\pi} \ln \frac{b}{a}.$$

$$B = \frac{\mu_0 I}{2\pi r},$$

$$U_m = \int_a^b \frac{B^2}{2\mu_0} 2\pi r l dr$$

$$= \int_a^b \frac{1}{2\mu_0} \left( \frac{\mu_0 I}{2\pi r} \right)^2 2\pi r l dr$$

$$= \frac{\mu_0 l I^2}{4\pi} \int_a^b \frac{dr}{r} = \frac{\mu_0 l I^2}{4\pi} \ln \frac{b}{a}.$$

**Mutual inductance**

When two loops not far from each other carry their currents $I_1$ and $I_2$ respectively, the fluxes through the loops are

the flux through loop 1 produced by the current on loop 2

$$\Phi_{1m} = \Phi_{11} + \underline{\Phi_{12}} = L_1 I_1 + \underline{L_{12} I_2}$$

$$\Phi_{2m} = \underline{\Phi_{21}} + \Phi_{22} = \underline{L_{21} I_1} + L_2 I_2$$

the flux through loop 2 produced by the current on loop 1

It can be proved that

$$L_{12} = L_{21} \equiv M. \qquad \text{---mutual inductance}$$

$$\varepsilon_{1i} = -\frac{d\Phi_{1m}}{dt} = -L_1 \frac{dI_1}{dt} - M \frac{dI_2}{dt},$$

$$\varepsilon_{2i} = -\frac{d\Phi_{2m}}{dt} = -M \frac{dI_1}{dt} - L_2 \frac{dI_2}{dt}.$$

In practice, mutual inductance can be determined by experiment.

The energy required to establish currents in the two circuits is

$$U_m = \int_0^t -\varepsilon_{1i} I_1 dt + \int_0^t -\varepsilon_{2i} I_2 dt$$

$$= \int_0^t \left( L_1 \frac{dI_1}{dt} + M \frac{dI_2}{dt} \right) I_1 dt + \int_0^t \left( M \frac{dI_1}{dt} + L_2 \frac{dI_2}{dt} \right) I_2 dt$$

$$= \int_0^{I_1} L_1 I_1 dI_1 + \int_0^{I_2} L_2 I_2 dI_2 + \int_0^{I_1 I_2} M d(I_1 I_2)$$

$$= \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2.$$

To generalize to the $k$ coupled circuits

$$U_m = \frac{1}{2}\sum_{i=1}^{k} L_i I_i^2 + \frac{1}{2}\sum_{\substack{i,j=1 \\ i\neq j}}^{k} M_{ij} I_i I_j .$$

two solenoids of turns $N_1$, $N_2$ are cross-winded

$$L_1 = \frac{N_1 \Phi_{11}}{I_1}, L_2 = \frac{N_2 \Phi_{22}}{I_2},$$

$$M = \frac{N_1 \Phi_{12}}{I_2} = \frac{N_2 \Phi_{21}}{I_1}.$$

$$\Phi_{11} = \Phi_{21}, \quad \Phi_{22} = \Phi_{12}.$$

$$M = \frac{N_1 \Phi_{22}}{I_2} = \frac{N_2 \Phi_{11}}{I_1}$$

$$M^2 = L_1 L_2, \quad M = \sqrt{L_1 L_2}.$$

In general

$$M = k\sqrt{L_1 L_2}, \qquad (0 \leq k \leq 1).$$

two $L_1$, $L_2$ connected in series and separated far enough

$$I_1 = I_2 = I,$$

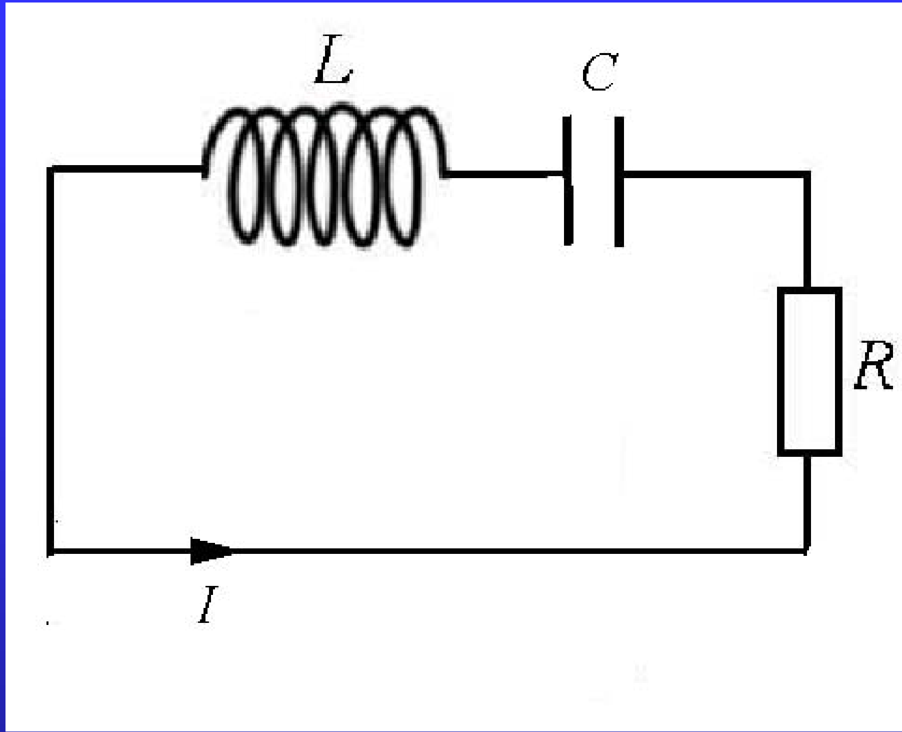
$$\mathcal{E}_i = \mathcal{E}_{1i} + \mathcal{E}_{2i} = -L_1 \frac{dI}{dt} - L_2 \frac{dI}{dt}$$

$$= -\left(L_1 + L_2\right)\frac{dI}{dt}, \quad \Rightarrow \quad L = L_1 + L_2.$$

# The *LC* oscillator

$$V_L = -L\frac{dI}{dt},$$

$$V_C = \frac{q}{C}$$




Applying Ohm's law, we have

$$RI = -L\frac{dI}{dt} - \frac{q}{C}.$$

Taking derivative of both sides of the equation gives

$$R\frac{dI}{dt} = -L\frac{d^2I}{dt^2} - \frac{1}{C}\underbrace{\frac{dq}{dt}}_{=I},$$

or

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}I = 0. \qquad (1)$$

When $R$=0, the above equation becomes

$$\frac{d^2I}{dt^2} + \frac{1}{LC}I = 0 \quad \text{or} \quad \frac{d^2I}{dt^2} + \omega_0^2 I = 0\,.$$

$$\omega_0 = \sqrt{\frac{1}{LC}}$$ the characteristic frequency of the circuit

$$I = I_0\cos(\omega_0 t + \varphi).$$

When $R \neq 0$ , the Eq.(1) can be rewritten as

$$\frac{d^2 I}{dt^2} + 2\gamma \frac{dI}{dt} + \omega_0^2 I = 0,$$

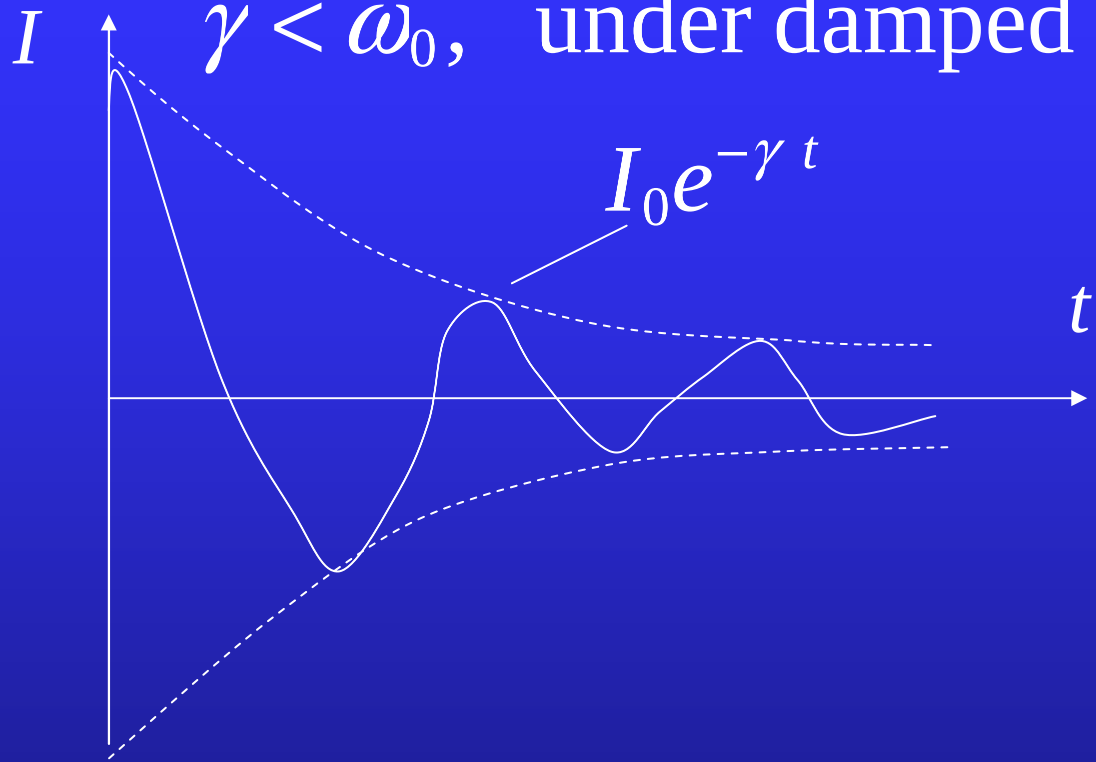
where $\gamma = R/2L$. It has the same form of the equation of motion for damped harmonic oscillation. Then, the solution is

$$\gamma < \omega_0, \quad I = I_0 e^{-\gamma t}\cos(\omega t + \varphi), \ \omega = \sqrt{\omega_0^2 - \gamma^2}.$$

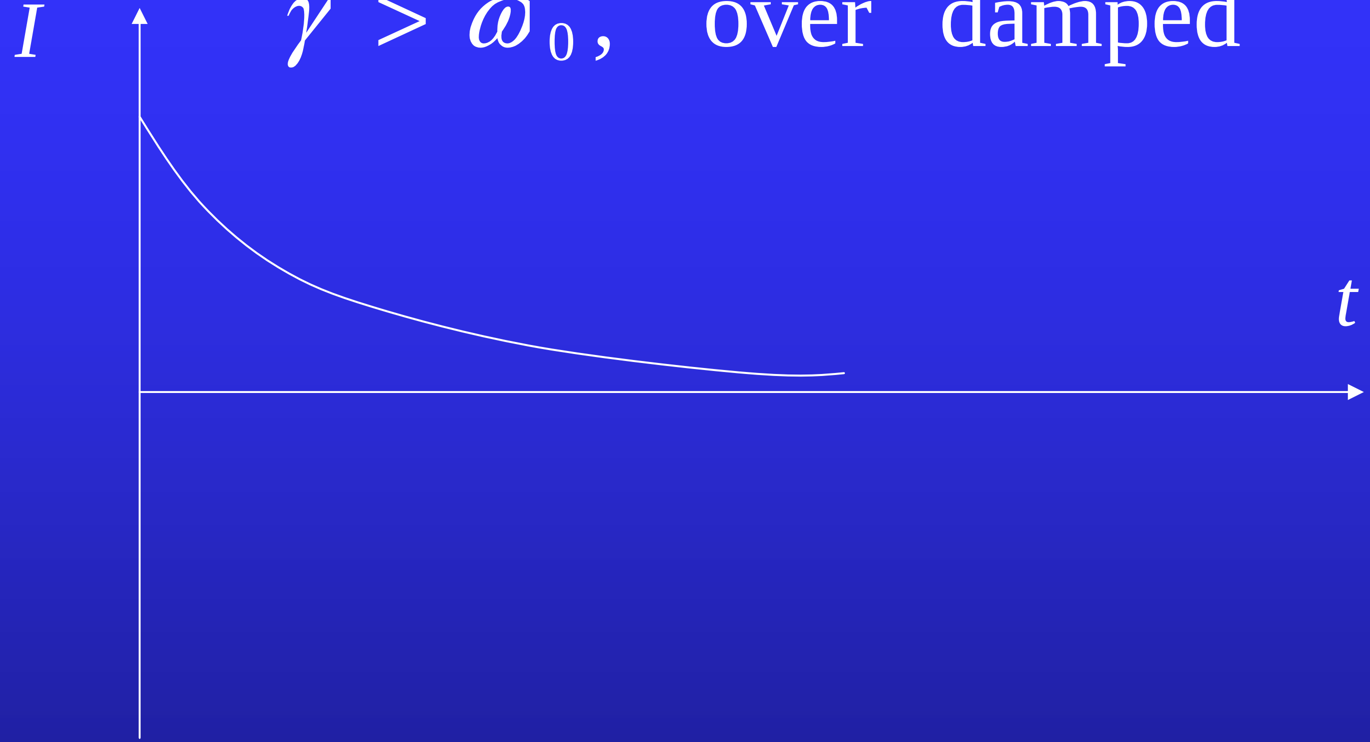
$$\gamma = \omega_0, \quad I = (I_0 + ct)e^{-\gamma t}.$$

$$\gamma > \omega_0, \quad I = I_0 e^{-\gamma t}[\cosh(\beta t) + c\sinh(\beta t)], \ \beta = \sqrt{\gamma^2 - \omega_0^2}.$$

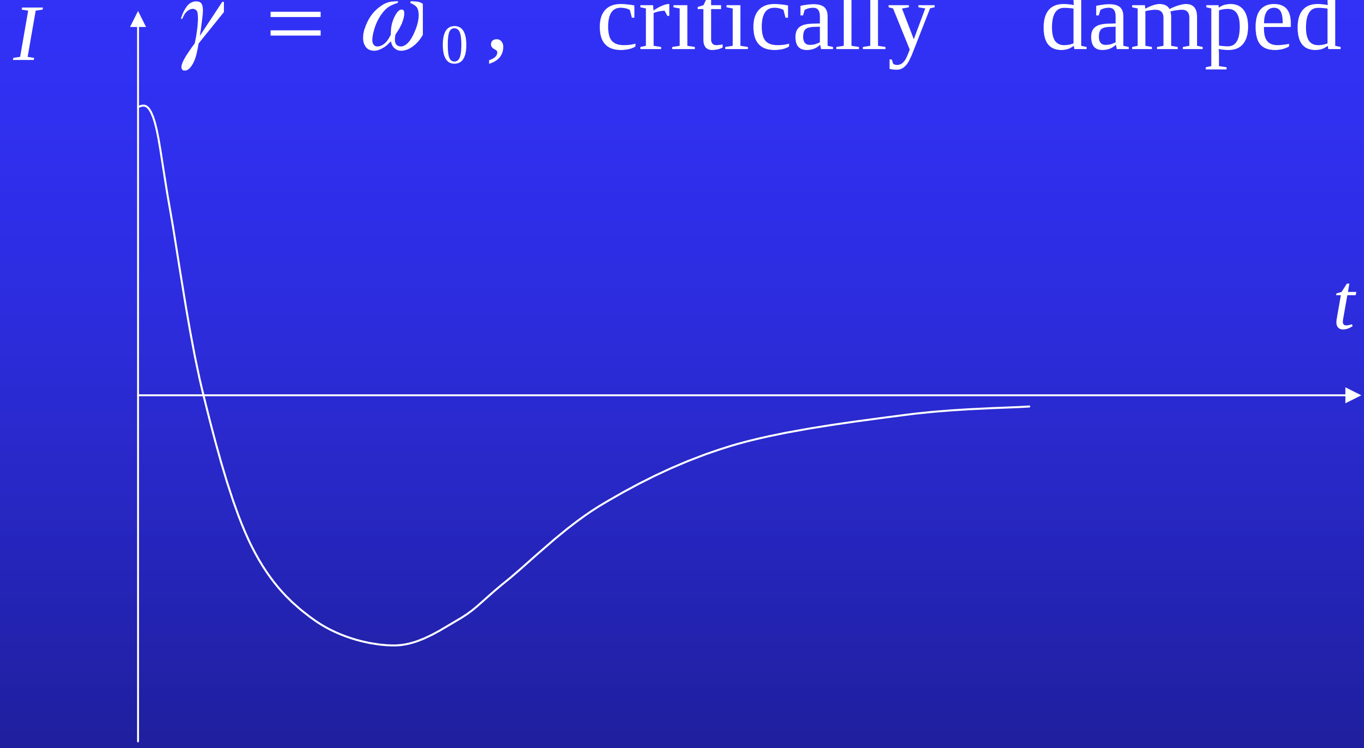
$\gamma < \omega_0$, under damped



$\gamma > \omega_0$ , over damped



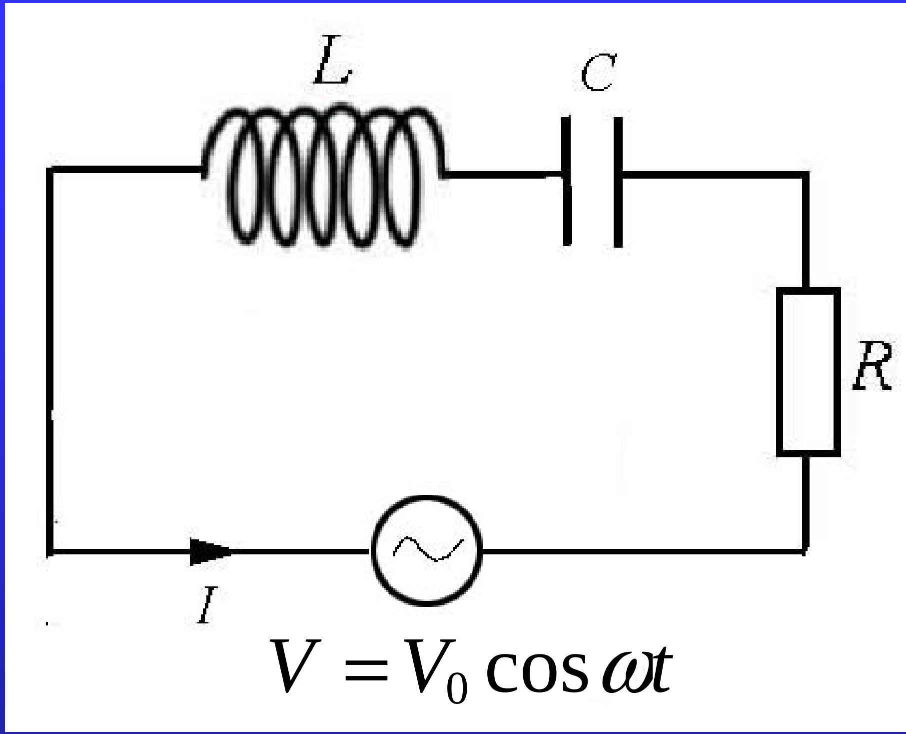
$\gamma = \omega_0$, critically damped



## Forced electrical oscillations

The differential equation of the current is

$$RI = -L\frac{dI}{dt} - \frac{q}{C} + V_0\cos\omega t.$$




Differentiating both sides with respect to time and rearranging the terms, we obtain

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{1}{C}I = -V_0\omega\sin\omega t. \qquad (1)$$

Eq.(1) can be rewritten as

$$\frac{d^2I}{dt^2} + 2\gamma\frac{dI}{dt} + \omega_0^2 I = \boxed{-\frac{V_0\omega}{L}\sin\omega t.} \qquad (2)$$

the inhomogeneous term

The solution can be written as

$$I(t) = I_h(t) + I_p(t).$$

the solution of the corresponding homogeneous equation

damped (transient process)

the particular solution

Assume

$$I(t) = I_0 \cos(\omega t - \alpha).$$

Substituting above expression into Eq.(2), we have

$$I_0[-\omega^2 \cos(\omega t - \alpha) - 2\gamma\omega \sin(\omega t - \alpha)$$

$$+\omega_0^2 \cos(\omega t - \alpha)] = -\frac{V_0 \omega}{L} \sin \omega t,$$

or

$$I_0[(\omega_0^2 - \omega^2)\cos\alpha + 2\gamma\omega \sin\alpha]\cos\omega t$$

$$+I_0[(\omega_0^2 - \omega^2)\sin\alpha - 2\gamma\omega\cos\alpha]\sin\omega t$$

$$= -\frac{V_0\omega}{L}\sin\omega t.$$

Comparing the coefficients of $\cos\omega t$ and $\sin\omega t$ respectively, we have

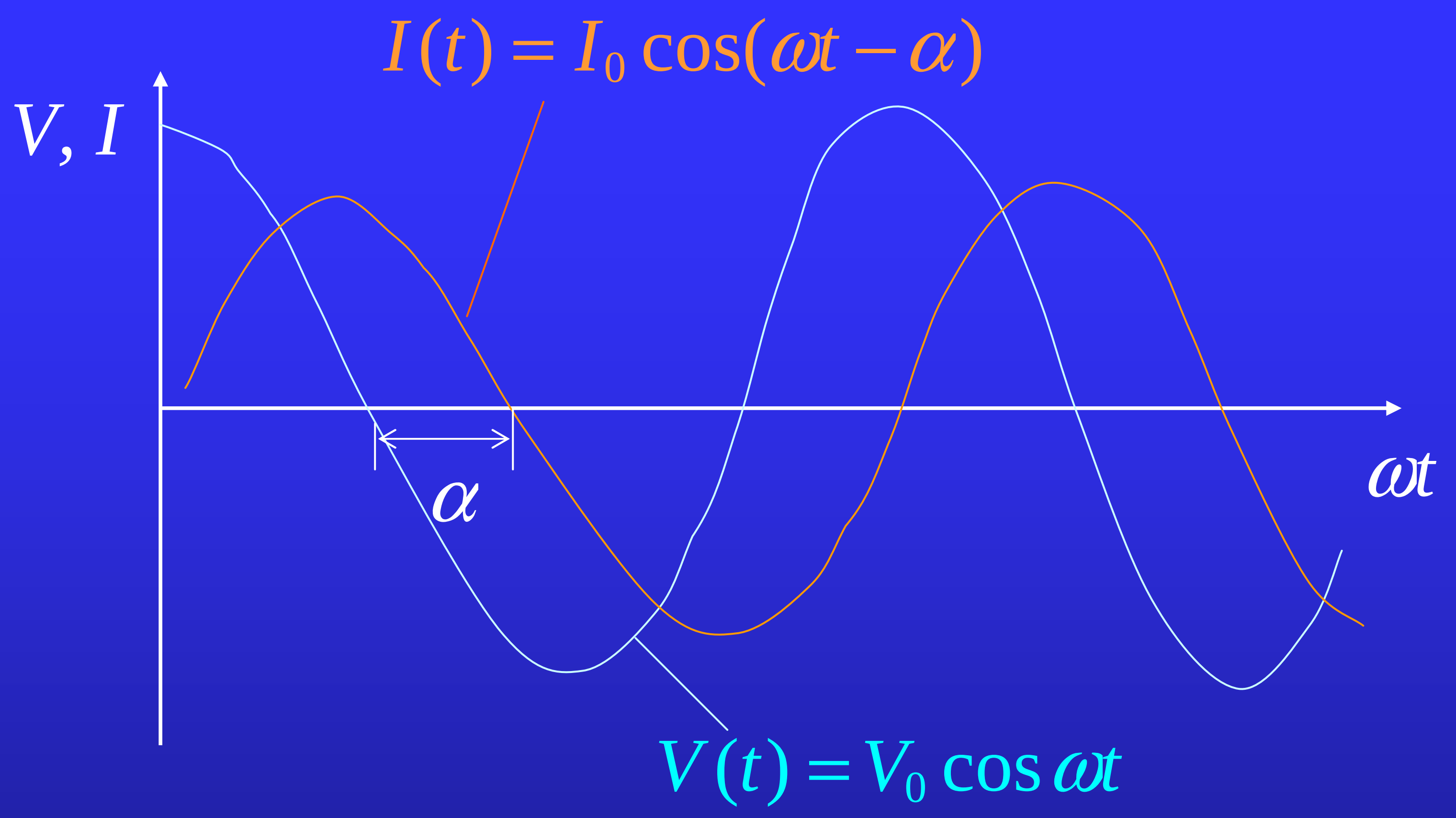
$$(\omega_0^2-\omega^2)\cos\alpha+2\gamma\omega\sin\alpha=0,$$

$$I_0[(\omega_0^2-\omega^2)\sin\alpha-2\gamma\omega\cos\alpha]=-\frac{V_0\omega}{L}.$$

Therefore,

$$\tan\alpha=\frac{\omega_0^2-\omega^2}{2\gamma\omega}=\frac{\omega L-\dfrac{1}{\omega C}}{R},$$

$$I_0=\frac{\dfrac{V_0}{L}}{\sqrt{(\omega_0^2-\omega^2)^2+4\gamma^2\omega^2}}=\frac{V_0}{\sqrt{R^2+\left(\omega L-\dfrac{1}{\omega C}\right)^2}}.$$




$\alpha$ ---the phase lag

$$\omega L - \frac{1}{\omega C} > 0, \qquad \alpha > 0;$$

$$\omega L - \frac{1}{\omega C} < 0, \qquad \alpha < 0.$$

$$C = 0, R = 0 \qquad \alpha = \pi/2$$

$$L = 0, R = 0 \qquad \alpha = -\pi/2$$

$$L = C = 0 \qquad \alpha = 0$$

The impedance: (阻抗)  $Z = \sqrt{R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2}$

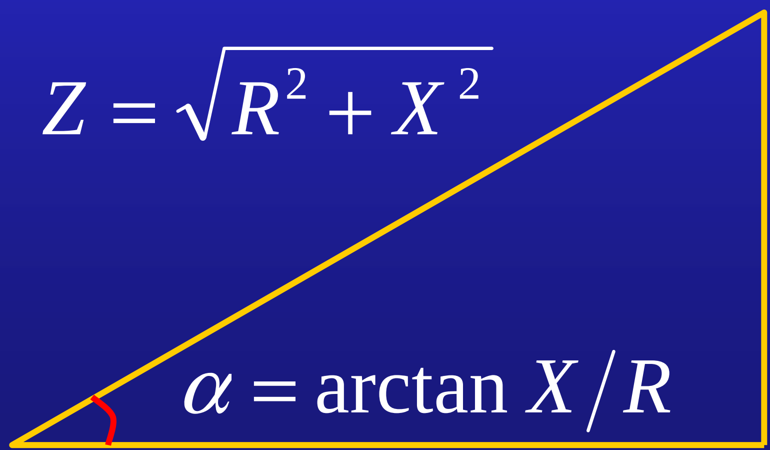
The resistance: (电阻)  $R$

The reactance: (电抗)  $X = \omega L - \dfrac{1}{\omega C}$

the inductive reactance(感抗)

the capacitive reactance(容抗)

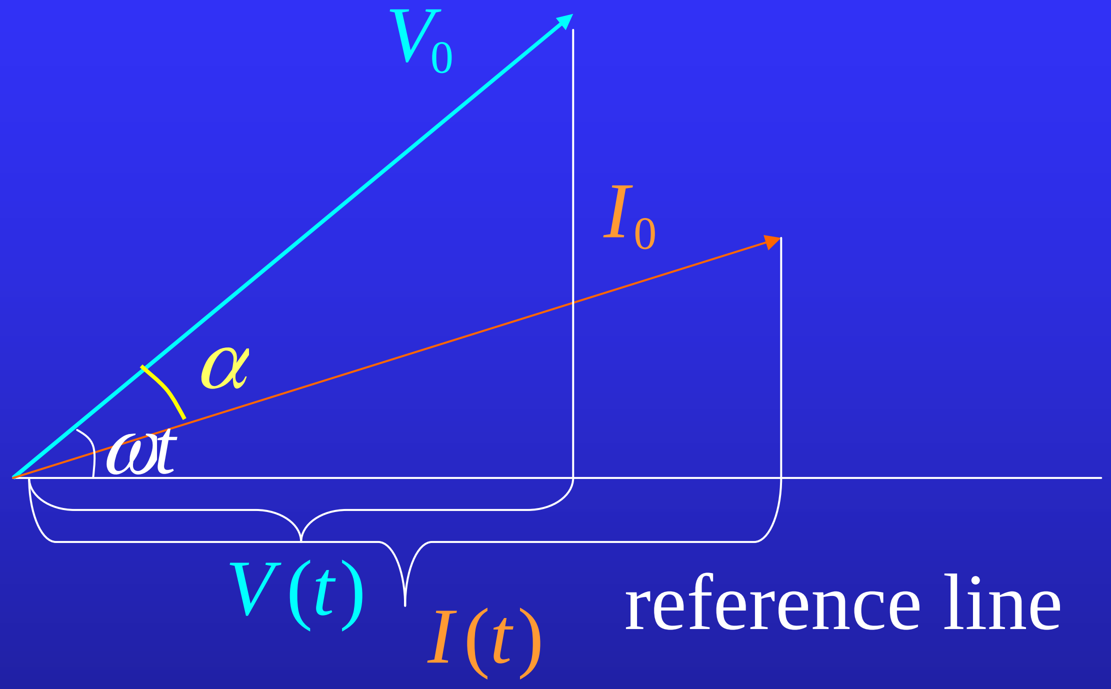
$$Z = \sqrt{R^2 + X^2}$$




$R$

$$X = \omega L - \frac{1}{\omega C}$$

# the rotating vector



<u>The electric power</u> at time $t$ is

$$P(t) = V(t)I(t) = V_0 I_0 \cos\omega t \cos(\omega t - \alpha)$$

$$= V_0 I_0 (\cos^2 \omega t \cos\alpha + \sin\omega t \cos\omega t \sin\alpha$$

$$= \frac{1}{2} V_0 I_0 [(1 + \cos 2\omega t)\cos\alpha + \sin 2\omega t \sin\alpha].$$

The average electric power over a single cycle is

$$P_{\text{ave}} = \frac{1}{T}\int_0^T P(t)dt = \frac{1}{2} V_0 I_0 \cos\alpha$$

$$= \frac{1}{2} I_0 Z I_0 \cos\alpha$$

$$= \frac{1}{2} R I_0^2.$$

$$I=\frac{I_0}{\sqrt{2}};\qquad V=\frac{V_0}{\sqrt{2}},\qquad \Rightarrow\qquad P_{\text{ave}}=VI\cos\alpha.$$

<u>the complex value representation</u>

$$V(t)=V_0\cos\omega t\qquad\rightarrow\qquad \tilde{V}=V_0e^{i\omega t};$$

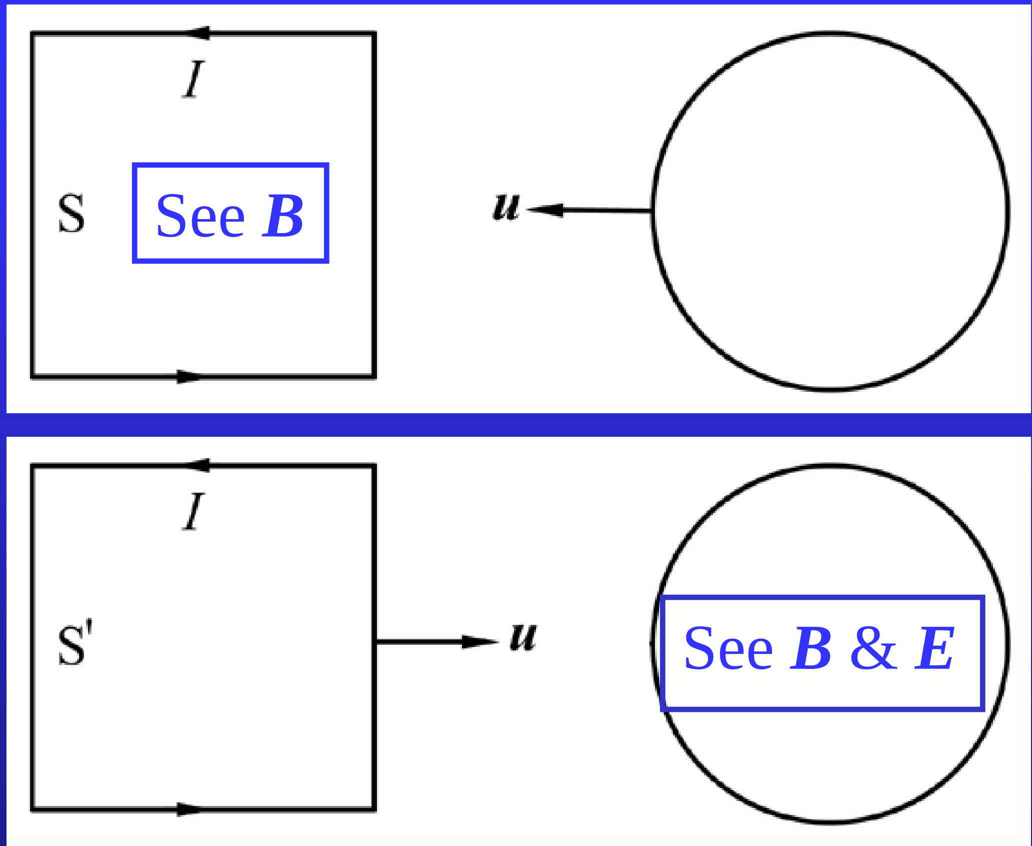
$$I(t)=I_0\cos(\omega t-\alpha)\qquad\rightarrow\qquad \tilde{I}=I_0e^{i(\omega t-\alpha)};$$

$$V(t)=\mathrm{Re}\,\tilde{V};\qquad I(t)=\mathrm{Re}\,\tilde{I}.$$

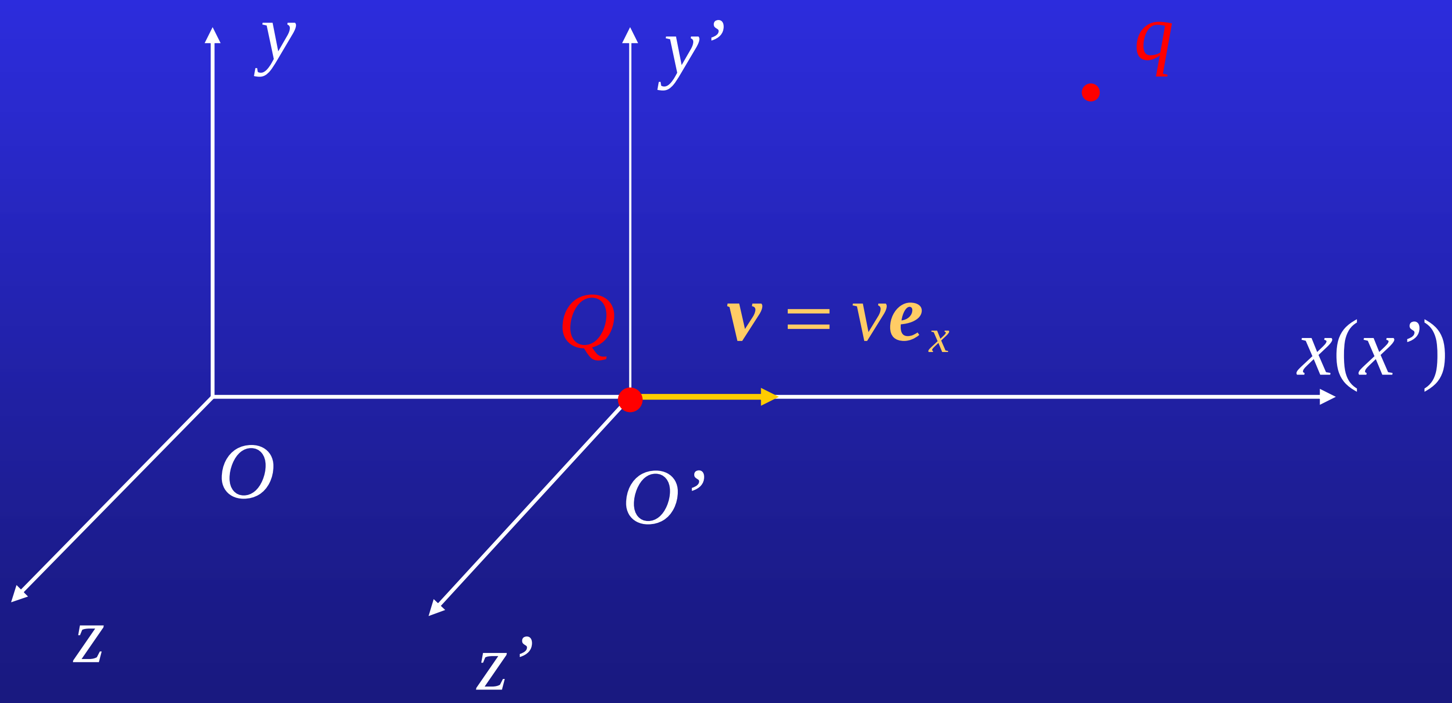
$$\tilde{Z}=Ze^{i\alpha}=R+i\left(\omega L-\frac{1}{\omega C}\right).$$

Ohm’s law: $\tilde{I}=\dfrac{\tilde{V}}{\tilde{Z}}.$

# 18.4 Relativity of *E* and *B*



Suppose two observers $O$ and $O'$ are in relative motion with velocity $\boldsymbol{v} = v\boldsymbol{e}_x$ and two charges $Q$ and $q$ are at rest relative to $O'$.



observer $O'$: only electric interaction between $Q$ and $q$.

$\boldsymbol{F}' = q\boldsymbol{E}'$. $\boldsymbol{E}'$---the electric field produced by $Q$ at the position of $q$.

The three components of the force are

$$F_x' = qE_x', \ F_y' = qE_y', \quad F_z' = qE_z'. \tag{1}$$

observer $O$: $Q$ is in motion. It produces both electric and magnetic field. $q$ is also in motion. The force exerted on $q$ is

$$\begin{aligned}\boldsymbol{F} &= q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \\ &= q\boldsymbol{E} + qv\boldsymbol{e}_x \times (B_x\boldsymbol{e}_x + B_y\boldsymbol{e}_y + B_z\boldsymbol{e}_z) \\ &= q\boldsymbol{E} + qvB_y\boldsymbol{e}_z - qvB_z\boldsymbol{e}_y.\end{aligned}$$

The three components of the force are

$$F_x = qE_x, \ F_y = q(E_y - vB_z), \quad F_z = q(E_z + vB_y). \tag{2}$$

The Lorentz transformation of the force is

$$F_x' = F_x, \quad F_y' = \frac{F_y}{\sqrt{1-\frac{v^2}{c^2}}}, \qquad F_z' = \frac{F_z}{\sqrt{1-\frac{v^2}{c^2}}}. \tag{3}$$

By comparing (1) and (2), we have

$$E_x' = E_x, \quad E_y' = \frac{E_y - vB_z}{\sqrt{1-\frac{v^2}{c^2}}}, \qquad E_z' = \frac{E_z + vB_y}{\sqrt{1-\frac{v^2}{c^2}}}. \tag{4}$$

The inverse transformation of (4) is

$$E_x = E'_x, \quad E_y = \frac{E'_y + vB'_z}{\sqrt{1-\frac{v^2}{c^2}}}, \quad E_z = \frac{E'_z - vB'_y}{\sqrt{1-\frac{v^2}{c^2}}}.$$

exchange the fields $\boldsymbol{E}, \boldsymbol{B} \leftrightarrow \boldsymbol{E}', \boldsymbol{B}'$

reverse the velocity $\boldsymbol{v} \to -\boldsymbol{v}$

It can be proved that

$$B'_x = B_x, \quad B'_y = \frac{B_y + \frac{vE_z}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad B'_z = \frac{B_z - \frac{vE_y}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}. \qquad (5)$$

The reverse transformation is

$$B_x = B'_x, \quad B_y = \frac{B'_y - \frac{vE'_z}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad B_z = \frac{B'_z + \frac{vE'_y}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}. \qquad (6)$$

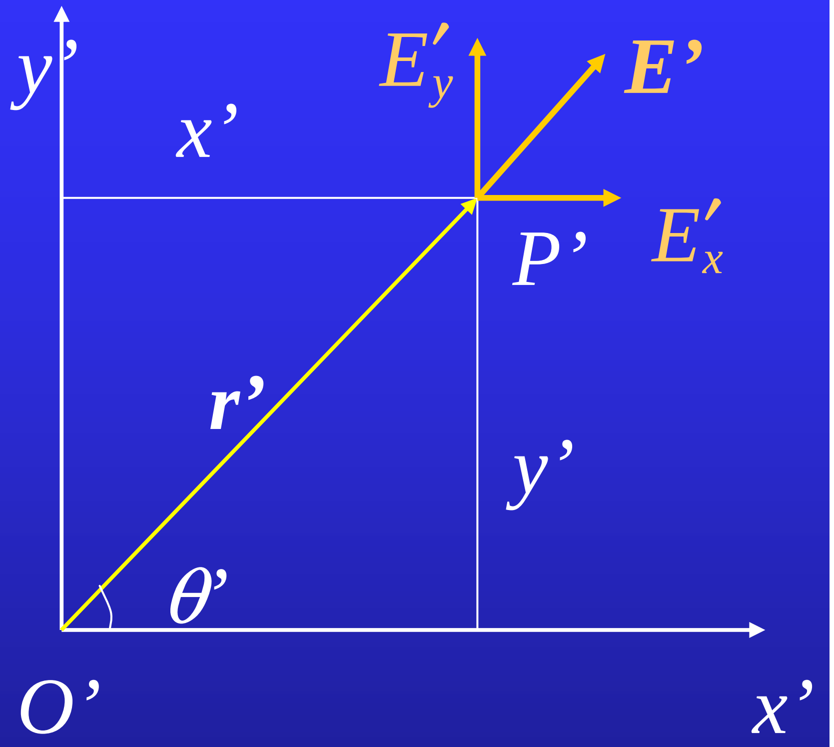
In general

$$\boldsymbol{E}' = \gamma\left[\boldsymbol{E} + c\boldsymbol{\beta}\times\boldsymbol{B}\right] - (\gamma - 1)\boldsymbol{e}_\beta(\boldsymbol{e}_\beta\cdot\boldsymbol{E})$$

$$\boldsymbol{B}' = \gamma\left[\boldsymbol{B} - \frac{1}{c}\boldsymbol{\beta}\times\boldsymbol{E}\right] - (\gamma - 1)\boldsymbol{e}_\beta(\boldsymbol{e}_\beta\cdot\boldsymbol{B})$$

**Example:** the electromagnetic field of a uniformly moving charge (relativistic).

**Solution:** Consider a charge q that is at rest relative to the frame *x'y'z'*. The frame *x'y'z'* is moving relative to *xyz* with a velocity *v* parallel to the *x*-axis. The observer *O'* measures no magnetic field but only electric field




$$\boldsymbol{E}' = \frac{q}{4\pi\varepsilon_0 r'^2}\boldsymbol{e}_{r'}. \qquad (1)$$

Then, using the Lorentz transformation for electromagnetic field, we have

$$E_x = E'_x,\ \ E_y = \frac{E'_y}{\sqrt{1-\frac{v^2}{c^2}}}, \quad E_z = \frac{E'_z}{\sqrt{1-\frac{v^2}{c^2}}}, \qquad (2)$$

and

$$B_x = 0,\ \ B_y = -\frac{vE_z}{c^2}, \quad B_z = \frac{vE_y}{c^2}, \qquad (3)$$

or

$$\boldsymbol{B} = \frac{\boldsymbol{v}\times\boldsymbol{E}}{c^2}. \qquad (4)$$

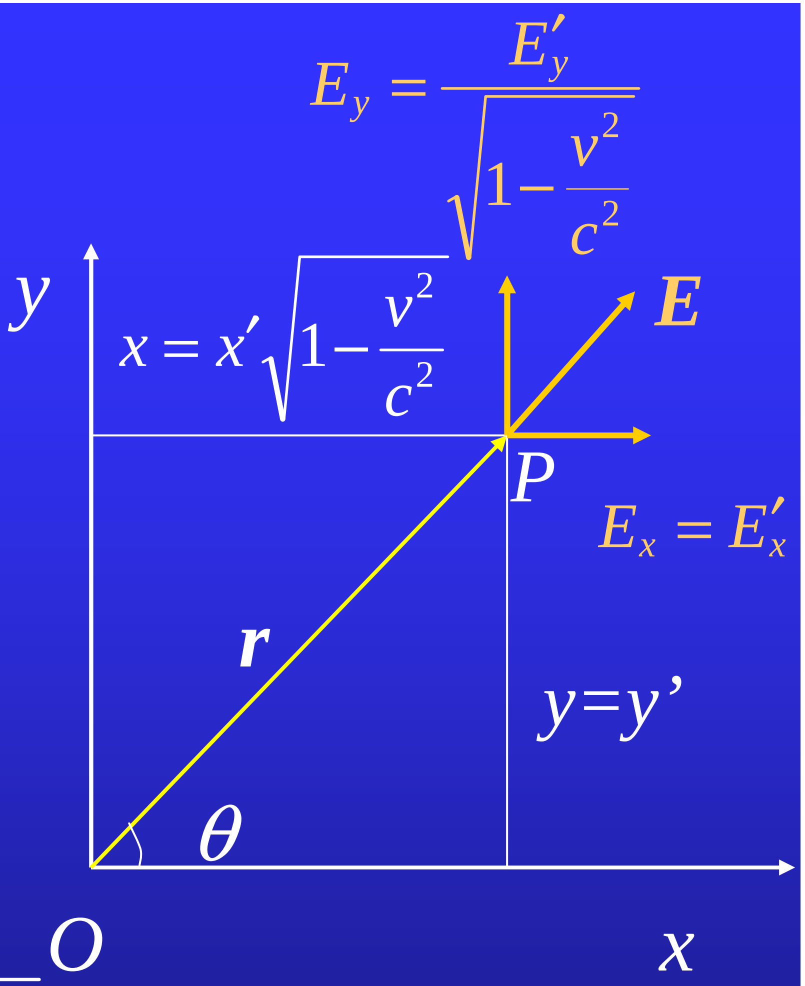
Using of the Lorentz contraction

$$x = x'\sqrt{1-\frac{v^2}{c^2}}, y = y', z = z',$$

and Eq.(2), we have

$$\boldsymbol{E} = \frac{q}{4\pi\varepsilon_0\sqrt{1-\frac{v^2}{c^2}}}\frac{x\boldsymbol{e}_x + y\boldsymbol{e}_y + z\boldsymbol{e}_z}{r'^3}$$

$$= \frac{q}{4\pi\varepsilon_0\sqrt{1-\frac{v^2}{c^2}}}\frac{x\boldsymbol{e}_x + y\boldsymbol{e}_y + z\boldsymbol{e}_z}{\left[\frac{x^2}{1-\frac{v^2}{c^2}} + y^2 + z^2\right]^{\frac{3}{2}}}$$



$$= \frac{q\left(1-\frac{v^2}{c^2}\right)}{4\pi\varepsilon_0} \frac{x\boldsymbol{e}_x + y\boldsymbol{e}_y + z\boldsymbol{e}_z}{\left[x^2 + \left(y^2 + z^2\right)\left(1-\frac{v^2}{c^2}\right)\right]^{\frac{3}{2}}}$$

$$= \frac{q\left(1-\frac{v^2}{c^2}\right)}{4\pi\varepsilon_0} \frac{x\boldsymbol{e}_x + y\boldsymbol{e}_y + z\boldsymbol{e}_z}{\left[r^2 - \frac{v^2}{c^2}(y^2 + z^2)\right]^{\frac{3}{2}}}$$

$$= \frac{q}{4\pi\varepsilon_0 r^2} \frac{1-\frac{v^2}{c^2}}{(1-\frac{v^2}{c^2}\sin^2\theta)^{\frac{3}{2}}} \boldsymbol{e}_r .$$

Thus, the magnetic field is

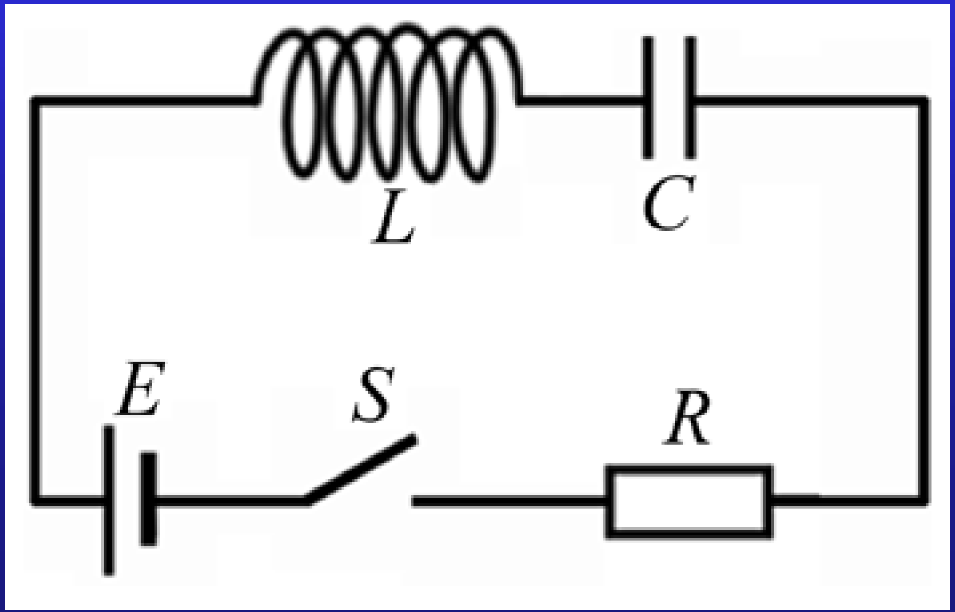
$$\boldsymbol{B} = \frac{q}{4\pi\varepsilon_0 c^2 r^2} \frac{1-\frac{v^2}{c^2}}{(1-\frac{v^2}{c^2}\sin^2\theta)^{\frac{3}{2}}} \boldsymbol{v} \times \boldsymbol{e}_r$$

$$= \frac{\mu_0 q}{4\pi r^2} \frac{1-\frac{v^2}{c^2}}{(1-\frac{v^2}{c^2}\sin^2\theta)^{\frac{3}{2}}} \boldsymbol{v} \times \boldsymbol{e}_r .$$

Here the relation $c^2 = \frac{1}{\varepsilon_0 \mu_0}$ has been used.

Assignment: 18.4; Supplementary 2; Supplementary 3

**Supplementary Problem 2**. Calculate the current in the circuit shown in the figure after the switch S is closed.



**Supplementary Problem 3.** Show that the resultant impedeanc of the circuit depicted in the figure is given by

$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\frac{1}{\omega L} - \omega C\right)^2}\,.$$